

Compound Poisson statistics for dynamical systems via spectral perturbation

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Abstract

We consider random transformations $T_\omega^n := T_{\sigma^{n-1}\omega} \circ \cdots \circ T_{\sigma\omega} \circ T_\omega$, where each map T_ω acts on a complete metrizable space M . The randomness comes from an invertible ergodic driving map $\sigma : \Omega \rightarrow \Omega$ acting on a probability space $(\Omega, \mathcal{F}, m)$. For a family of random target sets $H_{\omega,n} \subset M$ that shrink as $n \rightarrow \infty$, we consider quenched compound Poisson statistics of returns of random orbits to these random targets. We develop a spectral approach to such statistics: associated with the random map cocycle is a transfer operator cocycle $\mathcal{L}_{\omega,0}^n := \mathcal{L}_{\sigma^{n-1}\omega,0} \circ \cdots \circ \mathcal{L}_{\sigma\omega,0} \circ \mathcal{L}_{\omega,0}$, where $\mathcal{L}_{\omega,0}$ is the transfer operator for the map T_ω . We construct a perturbed cocycle with generator $\mathcal{L}_{\omega,n,s}(\cdot) := \mathcal{L}_{\omega,0}(\cdot e^{is\mathbb{1}_{H_{\omega,n}}})$ and an associated random variable $S_{\omega,n,k}(x) := \sum_{j=0}^{k-1} \mathbb{1}_{H_{\sigma^j\omega,n}}(T_\omega^j x)$, which counts the number of visits to random targets in an orbit of length k . Under suitable assumptions, we show that in the $n \rightarrow \infty$ limit, the random variables $S_{\omega,n,n}$ converge in distribution to a

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compound Poisson distributed random variable. We provide several explicit examples for piecewise monotone interval maps in both the deterministic and random settings.

Keywords: random dynamical systems, return-time statistics, extreme value theory, spectral perturbation, statistical properties of dynamical systems

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Contents

1. Introduction	2
1.1. Background and notation	3
1.2. Main result and literature review	4
2. The deterministic setting: a motivational approach	8
2.1. Further properties of the compound Poisson RV Z	11
2.2. Deterministic examples	12
3. The random setting: main results	20
3.1. Random perturbations	22
4. Checking our general assumptions for a large class of examples	35
5. Examples	40
Data availability statement	44
Acknowledgment	44
References	45

1. Introduction

In dynamical systems, return-time statistics describe the number of times that trajectories return to small neighbourhoods of the initial condition. In addition to their intrinsic interest, these statistics are heavily used in the study of extreme events. Indeed, the occurrence of such events can often be characterised as visits of trajectories of a dynamical system to a specific (small) region of the phase space, sometimes called a target set, or a hole.

This work investigates return-time statistics for various sufficiently chaotic dynamical systems, using compound Poisson random variables. These random variables describe the sum of a Poisson-distributed amount of independent identically-distributed random variables. More precisely, a random variable $Z : M \rightarrow \mathbb{N}$ is compound Poisson distributed if there exists a Poisson random variable N and a sequence $X_1, X_2, \dots$ of non-negative i.i.d. random variables $X_k : M \rightarrow \mathbb{N}$ such that

$$Z = \sum_{k=1}^N X_k. \quad (1.1)$$

Compound Poisson random variables are useful to model the cumulative effect of random events which occur in a fixed amount of time, e.g. the total amount of rainfall in a year.

A random dynamical system is defined through random compositions of maps $T_\omega : M \rightarrow M$, drawn from a collection $\{T_\omega\}_{\omega \in \Omega}$. An ergodic invertible measure-preserving map $\sigma : \Omega \rightarrow \Omega$ on a probability space $(\Omega, \mathcal{F}, m)$ creates a map cocycle, or random dynamical system, driven

by σ , namely $T_\omega^n := T_{\sigma^{n-1}\omega} \circ \cdots \circ T_{\sigma\omega} \circ T_\omega$. This very general driving setup enables the study of random dynamics far beyond (and including) deterministic maps (where $T_\omega = T$ for every $\omega \in \Omega$) and i.i.d. compositions of maps T_ω (where Ω is a set of bi-infinite sequences of symbols labelling the different maps, σ is the left shift and m is an infinite product measure $m = \nu^{\otimes \mathbb{Z}}$).

To investigate return-time statistics in the general setting of random dynamical systems, we consider a sequence of random target sets $H_{\omega,n} \subset M$, which are decreasing in $n \in \mathbb{N}$ for each $\omega \in \Omega$. For $\omega \in \Omega$, $x \in M$, $n \geq 1$, and $k \leq n$, we are interested in studying the following quantities

$$S_{\omega,n,k}(x) := \sum_{j=0}^{k-1} \mathbb{1}_{H_{\sigma^j \omega, n}}(T_\omega^j x),$$

which count the number of times an orbit of length k hits the random sequence of targets on their associated ω -fibres.

It is well known for deterministic dynamical systems, that the limit of $S_{\omega,n,k}$ for $n \rightarrow \infty$, is degenerate unless one chooses k as a function of the target set, which is known as a Kac scaling. In the context of random dynamical systems, there are different Kac-type scalings, see the discussion in remark 3.6. We will use the scaling (1.4) introduced in our former work [5], which means setting k to be $n = \left\lfloor \frac{t_\omega + \xi_{\omega,n}}{\mu_{\omega,0}(H_{\omega,n})} \right\rfloor$. We therefore consider the associated random variable $S_{\omega,n,n} : M \rightarrow \mathbb{N}$ on M and the discrete distributions

$$\mu_{\omega,0}(\{x \in M : S_{\omega,n,n}(x) = j\}), \quad j \geq 0 \quad (1.2)$$

in the limit $n \rightarrow \infty$, where $\{\mu_{\omega,0}\}_{\omega \in \Omega}$ is a family of probability measures on M satisfying the equivariance condition $\mu_{\omega,0} \circ T_\omega^{-1} = \mu_{\sigma\omega,0}$. These measures are a natural generalisation of invariant measures for a single deterministic map T to the setting of random dynamical systems. Roughly speaking, our main result (theorem 3.14) is that for m -a.e. ω , in the $n \rightarrow \infty$ limit, the random variable $S_{\omega,n,n}$ converges in distribution to a random variable Z that follows a compound Poisson distribution.

1.1. Background and notation

In our previous paper [5] we developed a spectral approach for a quenched extreme value theory that considers random dynamics on the unit interval with general ergodic invertible driving, and random observations. An extreme value law was derived using the first-order approximation of the leading Lyapunov multiplier of a suitably perturbed transfer operator defined by the introduction of small random holes in a metric space M . We were inspired by a result of Keller and Liverani [43] which, in the deterministic setting, developed abstract conditions on the transfer operator $\mathcal{L}$ and its perturbations $\mathcal{L}_\varepsilon$ (for each $\varepsilon > 0$) to ensure good first-order behaviour with respect to the perturbation size.

Our first task was to generalise the Keller–Liverani result when we have a sequential compositions of linear operators $\mathcal{L}_{\omega,0}^n := \mathcal{L}_{\sigma^{n-1}\omega,0} \circ \cdots \circ \mathcal{L}_{\sigma\omega,0} \circ \mathcal{L}_{\omega,0}$, where $\sigma : \Omega \rightarrow \Omega$ is an invertible, ergodic map on a configuration set Ω . We then consider a family of perturbed cocycles $\mathcal{L}_{\omega,\varepsilon}^n := \mathcal{L}_{\sigma^{n-1}\omega,\varepsilon} \circ \cdots \circ \mathcal{L}_{\sigma\omega,\varepsilon} \circ \mathcal{L}_{\omega,\varepsilon}$, for each $\varepsilon > 0$, where the size of the perturbation $\mathcal{L}_{\omega,0} - \mathcal{L}_{\omega,\varepsilon}$ is quantified by the value $\Delta_{\omega,\varepsilon} = \nu_{\sigma\omega,0}((\mathcal{L}_{\omega,0} - \mathcal{L}_{\omega,\varepsilon})(\phi_{\omega,0}))$, where $\phi_{\omega,0}$ and $\nu_{\omega,0}$ (the conformal measure), are respectively the random eigenvector of $\mathcal{L}_{\omega,0}$ and of its dual with common eigenvalue $\lambda_{\omega,0}$. We obtained an abstract quenched formula for the Lyapunov multipliers $\lambda_{\omega,0}$ up to first order in the size of the perturbation $\Delta_{\omega,\varepsilon}$. This map cocycle generates a transfer operator cocycle $\mathcal{L}_{\omega,0}^n$, where $\mathcal{L}_{\omega,0}$ is the transfer operator for the map T_ω . For each $\omega \in \Omega$ and each $\varepsilon > 0$, a random target, referred to as a hole in [5], $\tilde{H}_{\omega,\varepsilon} \subset M$

is introduced; this allows us to define the perturbed transfer operator $\mathcal{L}_{\omega,\varepsilon}$ for the open map T_ω and target $\tilde{H}_{\omega,\varepsilon}$, namely $\mathcal{L}_{\omega,\varepsilon}(f) = \mathcal{L}_\omega(\mathbb{1}_{M \setminus \tilde{H}_{\omega,\varepsilon}} f)$.

Suppose now $h_\omega : M \rightarrow \mathbb{R}$ is a continuous function for each ω and write $\bar{z}_\omega$ as its essential supremum. For any $z_{\omega,N} < \bar{z}_\omega$ we can now define the set

$$\mathcal{H}_{\omega,z_{\omega,N}} := \{x \in M : h_\omega(x) - z_{\omega,N} > 0\},$$

which could be identified as a target in the space M . The suffix N for the point $z_{\omega,N}$ means that we will now consider N such targets in order to study the *distribution of non-exceedances*:

$$\begin{aligned} \mu_{\omega,0}(\{x \in M : h_{\sigma^j \omega}(T_\omega^j(x)) \leq z_{\sigma^j \omega,N}, j = 0, \dots, N-1\}) \\ = \mu_{\omega,0}\left(\left\{x \in M : T_\omega^j(x) \notin \mathcal{H}_{\sigma^j \omega, z_{\sigma^j \omega,N}} \text{ for } j = 0, \dots, N-1\right\}\right) \end{aligned} \quad (1.3)$$

where $\mu_{\omega,0}$ is the equivariant measure for the unperturbed system. Moreover we will require an asymptotic behaviour for the targets of the type

$$\mu_{\omega,0}(\mathcal{H}_{\omega,z_{\omega,N}}) = (t_\omega + \xi_{\omega,N})/N, \quad (1.4)$$

for a.e. ω and each $N \geq 1$, where t_ω is a positive random variable and $\xi_{\omega,N}$ goes to zero when $N \rightarrow \infty$; see section 3 for more details. Following the spectral approach of [41], we proved in [5] that the distribution (1.3) behaves asymptotically as $\lambda_{\omega,\varepsilon_N}^N / \lambda_{\omega,0}^N$, where $\lambda_{\omega,\varepsilon_N}$ is the Lyapunov multiplier of $\mathcal{L}_{\omega,\varepsilon_N}$. Such a multiplier is obtained by the first order perturbation of $\lambda_{\omega,0}$ quoted above, and ultimately will produce the limit Gumbel's law

$$\begin{aligned} \lim_{N \rightarrow \infty} \mu_{\omega,0}(x \in M : h_{\sigma^j \omega}(T_\omega^j(x)) \leq z_{\sigma^j \omega,N}, j = 0, \dots, N-1) \\ = \exp\left(-\int_{\Omega} t_\omega \theta_{\omega,0} \, dm(\omega)\right), \end{aligned}$$

where the *extremal index* $\theta_{\omega,0}$ is given by the limit $\theta_{\omega,0} = \lim_{N \rightarrow \infty} \frac{\lambda_{\omega,0} - \lambda_{\omega,\varepsilon_N}}{\Delta_{\omega,\varepsilon_N}}$.

There is another equivalent interpretation of Gumbel's law in terms of hitting times that will be useful to introduce the main topic of this paper. Let us now consider a general family of random targets $\{H_{\omega,n}\}_{\omega \in \Omega}$, satisfying $H_{\omega,n'} \subseteq H_{\omega,n}$, $n' \geq n$, and $\lim_{n \rightarrow 0} \mu_{\omega,0}(H_{\omega,n}) = 0$. The first (random) hitting time to a target, starting at initial condition x and random configuration ω , is defined by:

$$\tau_{\omega,H_{\omega,n}}(x) := \inf\{k \geq 1, T_\omega^k(x) \in H_{\sigma^k \omega,n}\}.$$

Under the assumptions which allowed us to get Gumbel's law, in particular (1.4) $\mu_{\omega,0}(H_{\omega,n}) = \frac{t_\omega + \xi_{\omega,n}}{n}$, with $\xi_{\omega,n}$ going to zero when $n \rightarrow \infty$, we can also prove that

$$\lim_{n \rightarrow \infty} \mu_{\omega,0}(\tau_{\omega,H_{\omega,n}}(x) > t_\omega) = \exp\left(-\int_{\Omega} t_\omega \theta_{\omega,0} \, dm(\omega)\right).$$

This result suggests that the exponential law given by the extreme value distribution describes the time between successive events in a Poisson process.

1.2. Main result and literature review

Given $\omega \in \Omega$ and a sequence of random target sets $H_{\omega,n} \subset M$, we consider the random variables

$$S_{\omega,n,n}(x) := \sum_{j=0}^{n-1} \mathbb{1}_{H_{\sigma^j \omega,n}}(T_\omega^j x), \quad (1.5)$$

where the initial condition $x \in M$ is distributed according to $\mu_{\omega,0}$. The main result of this work is that for an appropriate choice of target sets, the distribution of the random variables (1.5) converge to a compound Poisson distribution as $n \rightarrow \infty$. In order to establish this result, the measure of the target sets should follow the scaling (1.4). Our proof relies on computing the characteristic function of the random variable (1.5) and showing that, as $n \rightarrow \infty$, this characteristic function converges pointwise to the characteristic function of a random variable which is discrete and infinitely divisible, and therefore compound Poisson distributed. Our main result could be summarised as follows. See theorem 3.14 for a detailed statement.

Theorem A. *For the random perturbed system introduced above and satisfying assumptions (C1)–(C8) and (S) (see section 3 for full details), we have that for each $s \in \mathbb{R}$ and m -a.e. $\omega \in \Omega$*

$$\lim_{n \rightarrow \infty} \mu_{\omega,0} (e^{isS_{\omega,n,n}}) = \lim_{n \rightarrow \infty} \frac{\lambda_{\omega,n,n}^n}{\lambda_{\omega,0}^n} = \exp \left(- (1 - e^{is}) \int_{\Omega} t_{\omega} \theta_{\omega}(s) \, dm(\omega) \right). \quad (1.6)$$

The quantity $\lambda_{\omega,0}$ is the top Lyapunov multiplier of the operator $\mathcal{L}_{\omega,0}$, while $\lambda_{\omega,n,n}$ is the top multiplier of a perturbed operator $\mathcal{L}_{\omega,n,n}(f) := \mathcal{L}_{\omega,0}(f \cdot e^{is\mathbb{1}_{H_{\omega,n}}})$. The quantities $\lambda_{\omega,0}^n$ and $\lambda_{\omega,n,n}^n$ are products of n such multipliers along the ω -orbit of length n . Notice that as soon as we have the limit characteristic function on the right hand side of (1.6), we could use Lévy's inversion formula to get the mass distribution of (1.5), namely

$$\lim_{n \rightarrow \infty} \mu_{\omega,0}(S_{\omega,n,n} = k) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T e^{-isk} \exp \left(- (1 - e^{is}) \int_{\Omega} t_{\omega} \theta_{\omega}(s) \, dm(\omega) \right) ds.$$

To obtain the limit (1.6) we again use our generalised version of the Keller–Liverani theorem, with the associated extremal index $\theta_{\omega}(s)$ which is given by

$$\lim_{n \rightarrow \infty} \frac{\lambda_{\omega,0} - \lambda_{\omega,n,n}}{\lambda_{\omega,0} \mu_{\omega,0}(H_{\omega,n})} = (1 - e^{is}) \theta_{\omega}(s).$$

It is not the first time that the Keller–Liverani perturbation theorem has been used to get the Poisson distribution for the number of visits in small sets. We quote here the result [58] which holds in the deterministic setting and which exhibits several differences when compared to our approach. First of all [58] computed the Laplace transform of the random variable counting the number of visits in a decreasing sequence of sets with bounded cylindrical lengths around a non-periodic point. In the limit of vanishing measure of the target set, the Laplace transform converges to the Laplace transform of the usual Poisson distribution, which from now on will be referred to as *standard Poisson*. This result does not cover the case when the target set is around a periodic point or when it has a different geometrical shape. Furthermore, even if the limiting Laplace transform could in principle be computed, it is not clear how to invert it in order to get the probability mass distribution.

In section 2 of this paper we will sketch how to apply our perturbation scheme to deterministic systems. In particular we obtain the following corollary of theorem 1.6 specialised to the setting of deterministic dynamical systems where we consider the space Ω to be a singleton, see theorem 2.1 for more details.

Corollary B. *Assume that $n\mu(H_n) \rightarrow t$ for some $0 < t < \infty$ and that conditions (A1)–(A7) of [43] hold for the operators $\mathcal{L}_{n,s}$ for each $s \in \mathbb{R}$ and all $n \in \mathbb{N}$ sufficiently large. Then we have*

$$\lim_{n \rightarrow \infty} \int e^{isS_{n,n}} h \, dx = e^{-\theta(s)(1-e^{is})t}. \quad (1.7)$$

We will show in example 2.4 that for the aforementioned case of the target set around periodic points, we will get, as expected, the Pólya–Aeppli distribution. We note that if the target

sets are balls around a point z , then we have a dichotomy regarding the convergence of the distribution of the number of visits for systems with a strong form of decay of correlations. Either z is periodic and in that case we have convergence to a Pólya–Aeppli distribution as in example 2.4, or z is not periodic and in that case we have convergence to a standard Poisson process in example 2.3, see [7] for a rigorous proof of this claim for systems which exhibit correlation decay with respect to L^1 observables. Combining these results we prove the following theorem which characterizes the distributions that can be obtained from targets centred at finitely many periodic and aperiodic points from distinct orbits, see theorem 2.6 for more details.

Theorem C. *Let $M \in \mathbb{N}$ and for each $1 \leq j \leq M$ and $n \in \mathbb{N}$ suppose $x_j \in H_n^{(j)}$ is simple (see section 2 for details) with $H_n = \cup_{j=1}^M H_n^{(j)}$. For $m \leq M$ and $1 \leq j \leq m$ suppose the following*

- (1) x_j is periodic with prime period r_j ,
- (2) x_j is a continuity point of h ,
- (3) DT^{r_j} is continuous at x_j and $\alpha_j = 1/|DT^{r_j}(x_j)|$,
- (4) $H_n^{(j)}$ is centred around x_j for each $n \in \mathbb{N}$.

Now for $m+1 \leq k \leq M$ suppose that x_k is aperiodic. Further suppose that $x_j \neq T^\ell(x_k)$ for any $1 \leq j \neq k \leq M$ and any $\ell \in \mathbb{N}$, and that the following limit exists

$$\lim_{n \rightarrow \infty} \frac{\mu(H_n^{(j)})}{\mu(H_n)} = p_j.$$

Then the random variable Z defined by the characteristic function given in (1.7) can be written as a sum of independent random variables

$$Z = W_0 + \sum_{j=1}^m W_j,$$

where W_0 is standard Poisson with parameter $\vartheta_0 = t \sum_{j=m+1}^M p_j$ and W_j is Pólya–Aeppli distributed with parameters $\rho_j = \alpha_j$ and $\vartheta_j = tp_j(1 - \alpha_j)$ for each $1 \leq j \leq m$.

Other compound Poisson distributions will emerge in the random setting, where the notion of periodicity is lost, and we will give a few examples of them in section 5. We will see that it is easier to construct examples that are not standard Poisson and this could be interesting for application to the real world, where the noise is a constituent of the environment, see [12] for an application to climate time series.

In the deterministic framework, besides [26, 27], two other recent papers developed compound Poisson statistics for general classes of dynamical systems, [31, 34]. Both papers aimed to compare a given probability measure—namely the distribution of the number of visits to a set—to a compound Poisson distribution. This will give an error for the total variation distance between the two distributions. Any compound Poisson distribution depends upon a set of parameters $\lambda_l, l \geq 1$. It has been shown in [34], that those parameters are related to another sequence $\alpha_l, l \geq 1$, which are the limits of the distribution of higher-order returns and therefore are in principle computable. Whenever those limits exist and the α_l verify a summability condition, the error term can be evaluated in two possible ways. In [34], inspired in part by [17], the error term is evaluated using a very general approximation theorem that allows one to measure how close a return-time distribution is to a compound binomial distribution, which

in the limit converges to a compound Poisson distribution. In [31] the error term is instead evaluated by an adaption of the classical Stein–Chen method [18]. The two approaches target different classes of dynamical systems. The approach of [34] is more geometric and is adapted to differentiable systems which are not necessarily exponentially mixing, whereas the approach of [31] is more devoted to symbolic and ϕ -mixing systems.

For quenched random dynamical systems, there are already two contributions proving convergence to a compound Poisson distribution. The first one, [25], is based on analogous results in the deterministic setting due to Freitas *et al* [26]. What is actually shown is the convergence of marked point processes for random dynamical systems given by fibred Lasota Yorke maps. The technique is quite different from our spectral approach, even though it uses the Laplace transform to establish the convergence of random measures. In [25], our threshold assumption (1.4) is replaced by the Hüsler type condition

$$\sum_{j=0}^{N-1} \mu_{\omega,0} (h_{\sigma^j \omega} (T_{\omega}^j(x)) > z_{\sigma^j \omega, N}) \rightarrow t, \quad (1.8)$$

which has been used to deal with non-stationary extreme value theory, see [25, 28, 39], and is proved under strong mixing requirements of the driving system guaranteeing at least polynomial decay of correlations for the marginal measure $\tilde{\mu} = \int \mu_{\omega,0} dm$. In contrast, in this present work we instead only need that the driving system be ergodic. Moreover the unperturbed transfer operator is defined with the usual geometric potential, forcing all of the fiberwise conformal measures to be equal to the Lebesgue measure. Finally, the only example completely treated gives convergence to a standard Poisson distribution. Recently another paper dealt with quenched Poisson processes for random subshifts of finite type, [22]. It is proved that hitting times to dynamically defined cylinders converge to a Poisson point process under the law of random equivariant measures with super-polynomial decay of correlations. The second recent paper on quenched systems [2], which became available after the completion of the present work, adapts the deterministic framework of [34] to the random setting. The approach of [2] is based on a probabilistic block-approximation, which does not require exponential decay of correlations of the equivariant measure $\mu_{\omega,0}$. Instead, [2] requires that the correlations decay with some polynomial rate together with the correlations of the marginal measure $\tilde{\mu}$, see above. The allowance of polynomial decay requires the stronger assumption that the driving of the base dynamical system $\sigma : \Omega \rightarrow \Omega$ must be mixing, while we only require ergodicity in the base. In addition, the threshold assumption of [2] fixes the time of observation by averaging over the measures of the target sets, and is therefore a type of annealed version of our scaling (1.4), see remark 3.6. The convergence to the limit distribution is obtained along subsequences that require a certain kind of summability.

The history of Poissonian distributions for the number of visits in small sets dates back to the seminal papers by Pitskel [54] and Hirata [37]. There have been several other contributions employing different techniques; we provide here a non-exhaustive list: [1, 16, 32, 33, 38, 44, 45, 59, 60]. A complementary approach to the statistics of the number of visits has been developed in the framework of extreme value theory, where it is more often called point process, or particular kinds of it as the marked point process; besides the papers quoted above [25, 26], see also [7, 27]. The distribution of the number of visits to vanishing balls has been studied for systems modelled by a Young tower in [17, 35, 36, 52, 56], and for uniformly hyperbolic systems in [6, 15]. Recurrence in billiards provided recently several new contributions; for planar billiards in [13, 14, 29, 51] and in [53] spatio-temporal Poisson processes were obtained by recording not only the successive times of visits to a set, but also the positions.

2. The deterministic setting: a motivational approach

While our results in the random setting, presented in section 3, imply the compound Poisson statistics for the deterministic setting, in order to motivate our results for random dynamical systems, we first give a sketch of our approach to obtain compound Poisson statistics in the deterministic setting via Keller–Liverani perturbation theory. We will also present a few general considerations that can be immediately translated to the random setting.

Suppose that $I \subseteq \mathbb{R}$ is a compact interval and that $T: I \rightarrow I$ is a piecewise smooth ($C^{1+\alpha}$), piecewise expanding map with finitely many branches. Let μ be an absolutely continuous invariant measure for T . Consider a decreasing sequence of (non-random) targets H_n shrinking to a finite set. We aim to compute the following distribution

$$\lim_{n \rightarrow \infty} \mu \left(\sum_{i=0}^{k-1} \mathbb{1}_{H_n}(T^i x) = j \right)$$

for $j = 0, 1, \dots$, where, as usual, we set $k = n$ and assume the scaling

$$n = \left\lfloor \frac{t}{\mu(H_n)} \right\rfloor. \quad (2.1)$$

Rather than following the approach of Zhang [58], who computed the moment generating function, we compute the characteristic function (CF) of the sum

$$S_{n,k}(x) := \sum_{i=0}^{k-1} \mathbb{1}_{H_n}(T^i x). \quad (2.2)$$

We show that it converges to the CF of a random variable (RV) Z , which has a compound Poisson distribution. We let $\mathcal{L}$ denote the Perron–Frobenius operator acting on the space BV of complex-valued bounded variation functions, and for each $s \in \mathbb{R} \setminus \{0\}$ and $n \in \mathbb{N}$ we define the perturbed operator

$$\mathcal{L}_{n,s}(f) = \mathcal{L}(e^{is\mathbb{1}_{H_n}} f).$$

Note that if $s = 0$, then $\mathcal{L}_{n,0} = \mathcal{L}$ for all $n \in \mathbb{N}$. Iterates of this operator are given by

$$\mathcal{L}_{n,s}^k(f) = \mathcal{L}^k(e^{isS_{n,k}} f)$$

for each $n \geq 1$. If h is the density of μ (with respect to Lebesgue), then

$$\int e^{isS_{n,k}} h \, dx = \int \mathcal{L}^k(e^{isS_{n,k}} h) \, dx = \int \mathcal{L}_{n,s}^k(h) \, dx.$$

As usual we suppose that the operators $\mathcal{L}$ and $\mathcal{L}_{n,s}$ are quasi-compact, and in particular,

$$\mathcal{L}_{n,s}(\cdot) = \lambda_{n,s} \nu_{n,s}(\cdot) \phi_{n,s} + Q_{n,s}(\cdot),$$

where the above objects are deterministic versions of the corresponding random objects in (C3). As the targets H_n shrink to a finite set, the operator $\mathcal{L}_{n,s}$ approaches $\mathcal{L}$. The triple norm difference between the two operators is given by

$$|||(\mathcal{L} - \mathcal{L}_{n,s})||| \leq |1 - e^{is}| \mu(H_n) \leq 2\mu(H_n). \quad (2.3)$$

In Keller–Liverani [43] they consider the following normalising quantity

$$\Delta_{n,s} = \int (\mathcal{L} - \mathcal{L}_{n,s}) h \, dx = (1 - e^{is}) \mu(H_n). \quad (2.4)$$

In [41, section 4], using the Keller–Liverani perturbation theory of [42], Keller shows that the assumptions of [43] hold (namely (A1)–(A6) in [43] or (1)–(6) in [41]) for a similar setting⁵. Thus, applying theorem 2.1 of [43], we have

$$\lim_{n \rightarrow \infty} \frac{1 - \lambda_{n,s}}{\Delta_{n,s}} = 1 - \sum_{k=0}^{\infty} q_k(s) =: \theta(s), \quad (2.5)$$

where

$$q_k(s) = \lim_{n \rightarrow \infty} \frac{1}{(1 - e^{is}) \mu(H_n)} \int (\mathcal{L} - \mathcal{L}_{n,s}) \mathcal{L}_{n,s}^k (\mathcal{L} - \mathcal{L}_{n,s})(h) \, dx, \quad (2.6)$$

and the limit in (2.6) is assumed to exist. Calculating q_k yields

$$\begin{aligned} q_k(s) &= \lim_{n \rightarrow \infty} \frac{1}{(1 - e^{is}) \mu(H_n)} \int (\mathcal{L} - \mathcal{L}_{n,s}) \mathcal{L}_{n,s}^k (\mathcal{L} - \mathcal{L}_{n,s})(h) \, dx \\ &= \lim_{n \rightarrow \infty} \frac{1}{(1 - e^{is}) \mu(H_n)} \int \left(1 - e^{is \mathbb{1}_{H_n}(T^{k+1}(x))}\right) e^{is[\mathbb{1}_{H_n}(T(x)) + \dots + \mathbb{1}_{H_n}(T^k(x))]} \\ &\quad \times \left(1 - e^{is \mathbb{1}_{H_n}(x)}\right) h(x) \, dx \\ &= \lim_{n \rightarrow \infty} \frac{1}{(1 - e^{is}) \mu(H_n)} \int_{H_n \cap T^{-(k+1)}(H_n)} (1 - e^{is})^2 e^{is[\mathbb{1}_{H_n}(T(x)) + \dots + \mathbb{1}_{H_n}(T^k(x))]} h(x) \, dx \\ &= \lim_{n \rightarrow \infty} \frac{(1 - e^{is})}{\mu(H_n)} \int_{H_n \cap T^{-(k+1)}(H_n)} e^{is[\mathbb{1}_{H_n}(T(x)) + \dots + \mathbb{1}_{H_n}(T^k(x))]} h(x) \, dx. \end{aligned} \quad (2.7)$$

In view of (2.7), we see that the orbit $T^\ell(x)$ can return to H_n for $\ell = 1, \dots, k$. Each return adds a multiplicative factor e^{is} , for a positive ℓ . Therefore, for $\ell = 0, \dots, k$, we define

$$\beta_n^{(k)}(\ell) := \frac{\mu\left(x; x \in H_n, T^{k+1}(x) \in H_n, \sum_{j=1}^k \mathbb{1}_{H_n}(T^j x) = \ell\right)}{\mu(H_n)}.$$

Now suppose that the limit

$$\beta_k(\ell) := \lim_{n \rightarrow \infty} \beta_n^{(k)}(\ell)$$

exists. Then we have the following alternate formulation of the q_k ,

$$q_k(s) = \lim_{n \rightarrow \infty} \frac{1}{1 - e^{is}} \sum_{\ell=0}^k (1 - e^{is})^2 e^{i\ell s} \beta_n^{(k)}(\ell) = (1 - e^{is}) \sum_{\ell=0}^k e^{i\ell s} \beta_k(\ell). \quad (2.8)$$

It follows from the assumption (A3) of [43] (the deterministic version of our assumption (C4)) that the sum

$$\Sigma := \sum_{k=0}^{\infty} \sum_{\ell=0}^k \beta_k(\ell) \quad (2.9)$$

⁵ In fact Keller shows the assumptions (A1)–(A6) of [43] (the deterministic versions of our assumptions (C1)–(C7)) hold for further examples of shift maps and higher dimensional maps for which our theory also applies.

converges absolutely⁶. Thus, combining (2.5) and (2.8) we obtain the following alternate expression for $\theta(s)$:

$$\theta(s) = 1 - (1 - e^{is}) \sum_{k=0}^{\infty} \sum_{\ell=0}^k e^{i\ell s} \beta_k(\ell). \quad (2.10)$$

In view of (2.5), and using (2.1) and (2.4), we have

$$1 - \lambda_{n,s} \approx \theta(s) \Delta_{n,s} = \theta(s) (1 - e^{is}) \frac{t}{n}.$$

Applying the approach of [41] to our perturbed operators $\mathcal{L}_{n,s}$, we obtain the following theorem.

Theorem 2.1. *Assume that the scaling (2.1) holds and that conditions (A1)–(A7) of [43] hold for the operators $\mathcal{L}_{n,s}$ for each $s \in \mathbb{R}$ and all $n \in \mathbb{N}$ sufficiently large. Then we have*

$$\lim_{n \rightarrow \infty} \int e^{is S_{n,n} h} dx = e^{-\theta(s)(1-e^{is})t} =: \varphi(s). \quad (2.11)$$

Taking (2.10) together with the absolute convergence of the series (2.9), we see that $\varphi(s)$ is continuous in $s = 0$, and therefore is the characteristic function of some random variable Z on some probability space $(\Gamma', \mathcal{B}', \mathbb{P}')$ to which the sequence of random variables (using (2.1) and (2.2))

$$Z_n := \sum_{i=0}^{\lfloor \frac{t}{\mu(H_n)} \rfloor} \mathbb{1}_{H_n} \circ T^i \quad (2.12)$$

converges in distribution; this follows from Lévy Continuity Theorem (see [47, theorem 3.6.1]). We let ν_Z denote the distribution of Z , and we now denote the invariant measure μ by $\mathbb{P}$. It follows from the Portmanteau Theorem that the variable Z is non-negative and integer valued as the distributional limit of a sequence of integer-valued RV⁷. Moreover it is a standard result that

$$\mathbb{P}'(Z = k) = \nu_Z(\{k\}) = \lim_{n \rightarrow \infty} \mathbb{P}(Z_n = k). \quad (2.13)$$

Note also that Z is clearly infinitely divisible, since $e^{-\theta(s)(1-e^{is})t} = (e^{-\theta(s)(1-e^{is})t/N})^N$. This along with the fact that Z is non-negative and integer valued imply that Z has a compound Poisson distribution, see for instance [40, p 389] or [24, section 12.2]. To obtain the probability mass function for Z we can apply the Lévy inversion formula to get

$$\nu_Z(\{k\}) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T e^{-isk} e^{-\theta(s)(1-e^{is})t} ds. \quad (2.14)$$

Recall the fact that $D_s^k \varphi(0) = i^k \mathbb{E}(Z^k)$ whenever the k th moments exists, where D_s denotes differentiation with respect to s . Thus, assuming that φ is twice differentiable, elementary

⁶ For more details in the random setting, see the arguments presented in section 3.

⁷ If ν_n denotes the distribution of Z_n , then the Portmanteau Theorem implies that $1 = \limsup_{n \rightarrow \infty} \nu_n(\mathbb{N}_0) = \nu_Z(\mathbb{N}_0)$, where $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$.

calculations⁸ give that

$$\mathbb{E}(Z) = t \quad \text{and} \quad \text{Var}(Z) = t(1 + 2\Sigma), \quad (2.15)$$

with Σ as in (2.9). If no confusion arises, we will denote the underlying probability with $\mathbb{P}$, instead of $\mathbb{P}'$, and its moments with $\mathbb{E}$, Var , etc which are actually computed with the distribution ν_Z .

2.1. Further properties of the compound Poisson RV Z

We now point out a few other properties enjoyed by compound Poisson random variables like our Z . First of all such a discrete random variable may be written as

$$Z := \sum_{j=1}^N X_j, \quad (2.16)$$

where the X_j are nonnegative, integer-valued, iid random variables defined on same probability space, and N is Poisson distributed with parameter ϑ . There is a simple relationship between the characteristic function $\varphi(s)$ of the variables Z and X_1 , namely we can write $\varphi(s) = \phi_Z(s)$ where $\phi_Z(s)$ is given by

$$\phi_Z(s) = e^{\vartheta(\phi_{X_1}(s)-1)}; \quad (2.17)$$

see [24, pp 287–288] for the analogous statement and argument for the moment generating function. Then we use the well-known fact that

$$\mathbb{P}(Z = k) = \sum_{\ell=0}^{\infty} \mathbb{P}(N = \ell) \mathbb{P}(S_{\ell} = k), \quad (2.18)$$

where $S_{\ell} = \sum_{i=1}^{\ell} X_i$, and S_0 is the random variable identically equal to 0. A standard result gives $\mathbb{E}(Z) = \vartheta \mathbb{E}(X_1)$, thus comparing with (2.15), we must have that $\vartheta = \frac{t}{\mathbb{E}(X_1)}$. Moreover it follows from the definition of the random variable Z_n (2.12) that (Gumbel's law):

$$\nu_Z(\{0\}) = \lim_{n \rightarrow \infty} \mu(Z_n = 0) = e^{-\theta_0 t},$$

where θ_0 is the extremal index, see e.g. [46].

If $k = 0$ in (2.18), then the only term which survives will be for $\ell = 0$, in which case we get

$$\mathbb{P}(Z = 0) = \mathbb{P}(N = 0) = e^{-\vartheta} = e^{-\frac{t}{\mathbb{E}(X_1)}},$$

and therefore,

$$\mathbb{E}(X_1) = \theta_0^{-1}.$$

This relationship was also obtained in [34], where a dynamical interpretation of the random variable X_j was given.

Furthermore, using the fact that $\vartheta = t\theta_0$, the fact that $\varphi(s) = \phi_Z(s)$, and (2.17), we get that

$$\phi_{X_1}(s) = \frac{\theta(s)(e^{is} - 1)}{\theta_0} + 1. \quad (2.19)$$

⁸ See section 3 for more details in the random setting.

Taking derivatives of ϕ_Z and ϕ_{X_1} we can additionally show⁹

$$\text{Var}(X_1) = \frac{\theta_0(1+2\Sigma) - 1}{\theta_0^2}.$$

Note that the probability generating function (PGF) for our compound Poisson RV Z is given by

$$G_Z(s) = e^{\vartheta(g_{X_1}(s)-1)}, \quad (2.20)$$

where $g_{X_1}(s)$ is the PGF for the RV X_1 . Set $\tilde{g}_{X_1} = \vartheta g_{X_1}$, and so we have $G_Z(s) = \exp(\tilde{g}_{X_1}(s) - \vartheta)$. Then Hoppe's form of the generalised chain rule applied to $G_Z(s)$ gives

$$D_s^K G_Z(s) = G_Z(s) \left(\sum_{k=0}^K \frac{(-1)^k}{k!} \sum_{j=0}^k (-1)^j \binom{k}{j} (\tilde{g}_{X_1}(s))^{k-j} D_s^K (\tilde{g}_{X_1}^j)(s) \right).$$

Then

$$\begin{aligned} D_s^K G_Z(0) &= G_Z(0) \left(\sum_{k=0}^K \frac{(-1)^k}{k!} \sum_{j=0}^k (-1)^j \binom{k}{j} (\tilde{g}_{X_1}(0))^{k-j} D_s^K (\tilde{g}_{X_1}^j)(0) \right) \\ &= e^{-t\theta_0} \left(\sum_{k=0}^K \frac{1}{k!} D_s^K (\tilde{g}_{X_1}^k)(0) \right) \end{aligned}$$

since $G_Z(0) = e^{-t\theta_0}$ and $g_{X_1}(0) = 0$. Since $\mathbb{P}(Z=K) = D_s^K G_Z(0)/K!$, we have

$$\mathbb{P}(Z=K) = \frac{e^{-t\theta_0}}{K!} \left(\sum_{k=0}^K \frac{1}{k!} D_s^K (\tilde{g}_{X_1}^k)(0) \right). \quad (2.21)$$

Remark 2.2. Note that $\mathbb{P}(Z=k)$ can be calculated using (2.13) or (2.14). To calculate $\mathbb{P}(X_1=k)$ we can apply the Lévy inversion formula to the CF ϕ_{X_1} to get

$$\mathbb{P}(X_1=k) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T e^{-isk} \left(\frac{\theta(s)(e^{is} - 1)}{\theta_0} + 1 \right) ds. \quad (2.22)$$

Note that the inversion formula can be used to numerically approximate the distribution of X_1 . Alternatively, we can recursively solve for $D_s^k \tilde{g}_{X_1}(0)$ (and thus $\mathbb{P}(X_1=k)$) in terms of $\mathbb{P}(Z=k)$ using (2.21).

2.2. Deterministic examples

We now present several examples in the interval map setting we have just described. We first show in examples 2.3 and 2.4 that if the targets H_n are centred around a single point x_0 , whose orbit has nonempty intersection with the set of discontinuities for T , then the return time distribution one can get is either standard Poisson (if x_0 is aperiodic) or Pólya–Aeppli (if x_0 is

⁹ See section 3 for details of the calculation in the random setting.

periodic). We then show in example 2.5 that if the targets H_n contain finitely many connected components which are centred around finitely many points, each of which has an orbit that is distinct from the others, then the return time distribution can be expressed as an independent sum of standard Poisson and Pólya–Aeppli random variables. Following this classification of the return time distributions for interval maps with targets centred around finitely many points with non-overlapping orbits, we give two examples, examples 2.7 and 2.8, describing the complicated behaviour that occurs when the targets are centred around points on the same orbit. These examples are related to clustering of extreme events created by multiple correlated maxima, investigated in [8, 19].

We will say that $x_0 \in I$ is *simple* if it is a point of continuity for T and if its forward orbit is disjoint from the discontinuity set of T . The following examples deal with targets centred at simple points. For instance, if T is smooth—not only piecewise smooth—then every $x_0 \in I$ is simple. For an analysis of extreme value laws associated to targets centred at non-simple points, we refer the reader to [7].

We begin with the simplest example of targets centred around an aperiodic point.

Example 2.3. Suppose that $x_0 \in H_n \subseteq I$, $n \geq 1$. If x_0 is simple and aperiodic, then using standard arguments with (2.7), it is easy to show that the $\beta_k(\ell)$ are zero for all $k \geq 0$, $0 \leq \ell \leq k$, and thus, the $q_k(s)$ are zero for all $k \geq 0$ and $s \in \mathbb{R}$. In this case we have that $\theta \equiv 1$, and so, using (2.1), the CF $\varphi(s) = e^{-t(1-e^{is})}$, which is the CF for a standard Poisson RV.

Example 2.4. Now suppose that $x_0 \in I$ is a simple periodic point of prime period $r \geq 1$, and that the targets H_n are centred around the point x_0 . Further suppose that the density h and DT^r are both continuous at x_0 . Again using standard arguments with (2.7), we see that all the $q_k = 0$ for $k \not\equiv r-1 \pmod{r}$ and that for $k = br-1$, $b \geq 1$, we have

$$q_{br-1}(s) = \lim_{n \rightarrow \infty} \frac{(1-e^{is})}{\mu(H_n)} \int_{H_n \cap T^{-br}(H_n)} e^{(b-1)is} h(x) dx = (1-e^{is}) e^{(b-1)is} \alpha^b,$$

since

$$\lim_{n \rightarrow \infty} \frac{\mu(H_n \cap T^{-br}(H_n))}{\mu(H_n)} = \alpha^b,$$

where $\alpha = 1/|DT^r(x_0)| < 1$. In this case θ becomes

$$\theta(s) = 1 - \sum_{b=1}^{\infty} q_{br-1}(s) = 1 - (1-e^{is}) \sum_{b=1}^{\infty} e^{(b-1)is} \alpha^b = 1 - \frac{1-e^{is}}{\alpha^{-1}-e^{is}} = \frac{1-\alpha}{1-\alpha e^{is}}.$$

Thus, using (2.1), the CF $\varphi(s)$ is given by

$$\varphi(s) = e^{-t(1-e^{is})\theta(s)} = e^{-t(1-e^{is})\left(\frac{1-\alpha}{1-\alpha e^{is}}\right)} = e^{-t(1-\alpha)\left(\frac{1-e^{is}}{1-\alpha e^{is}}\right)},$$

which is the CF of a Pólya–Aeppli distributed random variable Z with parameters $\rho = \alpha \in (0, 1)$ and $\vartheta = t(1-\alpha)$. In particular, this means that the random variables N and X_i in (2.16) are respectively Poisson with parameter ϑ and geometric with parameter ρ . The probability mass distribution of Z (see [40, section 9.6]) is given by

$$\mathbb{P}(Z=k) = \begin{cases} e^{-\vartheta} & \text{for } k=0, \\ e^{-\vartheta} \sum_{j=1}^k \frac{\vartheta^j}{j!} \rho^j (1-\rho)^{k-j} \binom{k-1}{j-1} & \text{for } k \geq 1. \end{cases}$$

We now present several examples describing the return times distributions one can obtain in the deterministic setting with targets with multiple connected components.

Example 2.5. Suppose $x_1, \dots, x_M \in I$ are simple and points of continuity for h , all of which belong to different orbits, so that $T^k(x_i) \neq x_j$ for any $i, j, k \in \mathbb{N}$ with $i \neq j$. Suppose that $H_n^{(j)}$ is an interval centred around the points x_j such that for all n sufficiently large we have $H_n^{(i)} \cap H_n^{(j)} = \emptyset$ for all $1 \leq i \neq j \leq M$. Set $H_n := \cup_{j=1}^M H_n^{(j)}$ and we assume the following limit exists:

$$p_j := \lim_{n \rightarrow \infty} \frac{\mu(H_n^{(j)})}{\mu(H_n)}.$$

Fixing $k \in \mathbb{N}$, since the x_j all belong to distinct orbits, for all $n \in \mathbb{N}$ sufficiently large we have that

$$\bigcup_{i \neq j} \left(H_n^{(i)} \cap T^{-k} \left(H_n^{(j)} \right) \right) = \emptyset.$$

Thus, for each $k \in \mathbb{N}$, for n sufficiently large we can write

$$H_n \cap T^{-k}(H_n) = \bigcup_{j=1}^M \left(H_n^{(j)} \cap T^{-k} \left(H_n^{(j)} \right) \right),$$

and so we can decompose the integral

$$\int_{H_n \cap T^{-k}(H_n)} f(y) \, dy = \sum_{j=1}^M \int_{H_n^{(j)} \cap T^{-k}(H_n^{(j)})} f(y) \, dy.$$

Thus, we denote

$$I_{k,j}(s) := \lim_{n \rightarrow \infty} \frac{1 - e^{is}}{\mu(H_n)} \int_{H_n^{(j)} \cap T^{-(k+1)}(H_n^{(j)})} e^{is S_{n,k}(T(y))} h(y) \, dy.$$

which implies that $q_k(s) = \sum_{j=1}^M I_{k,j}(s)$. Suppose that x_j is periodic with prime period r_j and that DT^{r_j} is continuous at x_j for $1 \leq j \leq m$ (for $m \leq M$). For each $1 \leq j \leq m$ denote

$$\alpha_j = 1/|DT^{r_j}(x_j)|.$$

For $m+1 \leq j \leq M$ suppose that x_j is aperiodic. Applying the same arguments from both examples above we obtain

$$I_{k,j}(s) = \begin{cases} (1 - e^{is}) (e^{is(b-1)} p_j \alpha_j^b) & \text{if } x_j \text{ is periodic and } k = br_j - 1, \quad b \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

We set

$$\kappa_j(s) := \sum_{k=0}^{\infty} I_{k,j}(s),$$

and thus, if x_j is aperiodic ($m+1 \leq j \leq M$), we have $\kappa_j(s) \equiv 0$, and if x_j is periodic ($1 \leq j \leq m$), then following the calculation from the previous examples gives that

$$\kappa_j(s) = \frac{p_j \alpha_j (1 - e^{is})}{1 - \alpha_j e^{is}}.$$

In order to calculate the characteristic function $\varphi(s)$ for the random variable Z coming from (2.11), we first calculate $\theta(s)$. Collecting together the above calculations we can write

$$\begin{aligned}\theta(s) &= 1 - \sum_{k=0}^{\infty} q_k(s) = 1 - \sum_{j=1}^M \sum_{k=0}^{\infty} I_{k,j}(s) = 1 - \sum_{j=1}^M \kappa_j(s) = 1 - \sum_{j=1}^m \frac{p_j \alpha_j (1 - e^{is})}{1 - \alpha_j e^{is}} \\ &= \sum_{j=1}^M p_j - \sum_{j=1}^m \frac{p_j \alpha_j (1 - e^{is})}{1 - \alpha_j e^{is}} = \sum_{j=1}^m \frac{p_j (1 - \alpha_j)}{1 - \alpha_j e^{is}} + \sum_{j=m+1}^M p_j.\end{aligned}$$

Thus the CF $\varphi(s)$ is given by

$$\begin{aligned}\varphi(s) &= \exp \left(-t(1 - e^{is}) \sum_{j=1}^m \frac{p_j (1 - \alpha_j)}{1 - \alpha_j e^{is}} \right) \cdot \exp \left(-t(1 - e^{is}) \sum_{j=m+1}^M p_j \right) \\ &= \exp \left(-t(1 - e^{is}) \sum_{j=m+1}^M p_j \right) \cdot \prod_{j=1}^m \exp \left(-tp_j (1 - \alpha_j) \frac{1 - e^{is}}{1 - \alpha_j e^{is}} \right).\end{aligned}$$

Thus the random variable Z can be written as a sum of independent random variables $Z = W_0 + \sum_{j=1}^m W_j$ where W_0 is Poisson with parameter $\vartheta = t \sum_{j=m+1}^M p_j$ and W_j is Pólya–Aeppli distributed with parameters $\rho_j = \alpha_j$ and $\vartheta_j = tp_j(1 - \alpha_j)$ for each $1 \leq j \leq m$.

We have thus proved the following theorem classifying the full extent of the return time distributions when the targets are centred around finitely many simple points with distinct orbits.

Theorem 2.6. *Let $M \in \mathbb{N}$ and for each $1 \leq j \leq M$ and $n \in \mathbb{N}$ suppose $x_j \in H_n^{(j)}$ is simple with $H_n = \cup_{j=1}^M H_n^{(j)}$. For $m \leq M$ and $1 \leq j \leq m$ suppose the following*

- (1) x_j is periodic with prime period r_j ,
- (2) x_j is a continuity point of h ,
- (3) DT^{r_j} is continuous at x_j and $\alpha_j = 1/|DT^{r_j}(x_j)|$,
- (4) $H_n^{(j)}$ is centred around x_j for each $n \in \mathbb{N}$.

Now for $m+1 \leq k \leq M$ suppose that x_k is aperiodic. Further suppose that $x_j \neq T^\ell(x_k)$ for any $1 \leq j \neq k \leq M$ and any $\ell \in \mathbb{N}$, and that the following limit exists

$$\lim_{n \rightarrow \infty} \frac{\mu(H_n^{(j)})}{\mu(H_n)} = p_j.$$

Then the RV Z defined by the CF $\varphi(s)$ given in (2.11) can be written as a sum of independent random variables

$$Z = W_0 + \sum_{j=1}^m W_j,$$

where W_0 is standard Poisson with parameter $\vartheta_0 = t \sum_{j=m+1}^M p_j$ and W_j is Pólya–Aeppli distributed with parameters $\rho_j = \alpha_j$ and $\vartheta_j = tp_j(1 - \alpha_j)$ for each $1 \leq j \leq m$.

We now give examples that show the complexity that arises when the targets H_n are centred around multiple points with overlapping orbits.

Example 2.7. Suppose $x_1 \in I$ is aperiodic. Let $x_2 = T^b(x_1)$ for some $b \geq 1$. Suppose that x_1 (and thus also x_2) is simple, and suppose $H_n = H_n^{(1)} \cup H_n^{(2)}$ with $H_n^{(1)} \cap H_n^{(2)} = \emptyset$ and $H_n^{(j)}$ is centred around x_j for $j = 1, 2$ and each $n \in \mathbb{N}$. Further suppose the following limit exists:

$$p_j := \lim_{n \rightarrow \infty} \frac{\mu(H_n^{(j)})}{\mu(H_n)}$$

for $j = 1, 2$, and that h and DT^b are continuous at x_1 . Then all of the q_k are equal to zero except for q_{b-1} which is given by

$$\begin{aligned} q_{b-1}(s) &= \lim_{n \rightarrow \infty} \frac{1 - e^{is}}{\mu(H_n)} \int_{H_n \cap T^{-b}(H_n)} e^{is[\mathbb{1}_{H_n}(T(y)) + \dots + \mathbb{1}_{H_n}(T^{b-1}(y))]} h(y) \, dy \\ &= \lim_{n \rightarrow \infty} \frac{1 - e^{is}}{\mu(H_n)} \int_{H_n^{(1)} \cap T^{-b}(H_n^{(2)})} h(y) \, dy = (1 - e^{is}) \min\{p_1, p_2\alpha\}, \end{aligned} \quad (2.23)$$

where

$$\alpha = \frac{1}{|DT^b(x_1)|}.$$

Setting $\Gamma = \min\{p_1, p_2\alpha\}$ ¹⁰, then $\theta(s) = 1 - q_{b-1}(s) = 1 + (e^{is} - 1)\Gamma$, and so the CF $\varphi(s)$ from (2.11) is given by

$$\varphi(s) = \exp(t(e^{is} - 1)(1 + (e^{is} - 1)\Gamma)).$$

Note that φ is the CF for a RV Z whose distribution is compound Poisson, which is neither standard Poisson nor Pólya–Aeppli. In view of (2.17) and (2.19), we see that the CF of X_1 is given by

$$\phi_{X_1}(s) = 1 + \frac{(e^{is} - 1)(1 + \Gamma(e^{is} - 1))}{\Gamma},$$

where we have that $\theta_0 = \Gamma$ follows from similar calculations as (2.23); see [5].

Example 2.8. Suppose $x_1 \in I$ is a periodic point with prime period $r > 1$. Let $x_2 = T^b(x_1)$ for some $1 \leq b < r$. Suppose that x_1 is simple, and consequently so is x_2 . Further suppose that x_1 and x_2 are points of continuity for h , and suppose $H_n = H_n^{(1)} \cup H_n^{(2)}$ with $H_n^{(1)} \cap H_n^{(2)} = \emptyset$ and $H_n^{(j)}$ is centred around x_j for $j = 1, 2$ and each $n \in \mathbb{N}$. Further suppose the following limit exists:

$$p_j := \lim_{n \rightarrow \infty} \frac{\mu(H_n^{(j)})}{\mu(H_n)}$$

¹⁰ Note that $\Gamma = p_1$ when $H_n^{(1)} \subseteq T^{-b}(H_n^{(2)})$, and $\Gamma = p_2\alpha$ otherwise.

for $j = 1, 2$ and that DT^r is continuous at x_1 . Set

$$\alpha := \frac{1}{|DT^r(x_1)|} = \frac{1}{|DT^r(x_2)|},$$

and set

$$\gamma_1 := \frac{1}{|DT^{r-b}(x_2)|} \quad \text{and} \quad \gamma_2 := \frac{1}{|DT^b(x_1)|}.$$

As previously argued, we have

$$\lim_{n \rightarrow \infty} \frac{\mu(H_n^{(j)} \cap T^{-k}(H_n^{(j)}))}{\mu(H_n)} = p_j \alpha^{\frac{k}{r}}$$

for $j = 1, 2$ and $k \equiv 0 \pmod{r}$. Now if $k = ar + b$ then we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\mu(H_n^{(1)} \cap T^{-k}(H_n^{(2)}))}{\mu(H_n)} &= \lim_{n \rightarrow \infty} \frac{\min\left\{\mu(H_n^{(1)}), \mu(H_n^{(2)}) \frac{1}{|DT^{ar+b}(x_1)|}\right\}}{\mu(H_n)} \\ &= \min\left\{p_1, p_2 \alpha^a \cdot \frac{1}{|DT^b(x_1)|}\right\} =: \Gamma_{1,2}(a) \end{aligned}$$

and similarly for $k = ar - b$ we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\mu(H_n^{(2)} \cap T^{-k}(H_n^{(1)}))}{\mu(H_n)} &= \lim_{n \rightarrow \infty} \frac{\min\left\{\mu(H_n^{(2)}), \mu(H_n^{(1)}) \frac{1}{|DT^{ar-b}(x_2)|}\right\}}{\mu(H_n)} \\ &= \min\left\{p_2, p_1 \alpha^{a-1} \cdot \frac{1}{|DT^{r-b}(x_2)|}\right\} =: \Gamma_{2,1}(a). \end{aligned}$$

Note that there exist $a_{1,2}, a_{2,1} \geq 1$ such that $\Gamma_{1,2}(a) = p_2 \alpha^a \gamma_2$ for all $a \geq a_{1,2}$ and $\Gamma_{2,1}(a) = p_1 \alpha^{a-1} \gamma_1$ for all $a \geq a_{2,1}$. For each $k, j, i \in \mathbb{N}$ we set

$$I_{k,j,i}(s) := \lim_{n \rightarrow \infty} \frac{1 - e^{is}}{\mu(H_n)} \int_{H_n^{(j)} \cap T^{-(k+1)}(H_n^{(i)})} e^{is[\mathbb{1}_{H_n}(T(y)) + \dots + \mathbb{1}_{H_n}(T^k(y))]} h(y) dy$$

and hence for each k we have

$$q_k(s) := I_{k,1,1}(s) + I_{k,2,2}(s) + I_{k,1,2}(s) + I_{k,2,1}(s).$$

Following the previous arguments, for $k = ar - 1$ we have that

$$I_{ar-1,j,j}(s) = (1 - e^{is}) \left(e^{is(a-1)} p_j \alpha^a \right)$$

for $j = 1, 2$, and $I_{ar-1,1,2}(s) = I_{ar-1,2,1}(s) = 0$, so

$$\begin{aligned} I_{ar-1}(s) &= I_{ar-1,1,1}(s) + I_{ar-1,2,2}(s) \\ &= (1 - e^{is}) \left(e^{is(a-1)} (p_1 \alpha^a + p_2 \alpha^a) \right) = (1 - e^{is}) \left(e^{is(a-1)} \alpha^a \right). \end{aligned}$$

Now for $k = ar + b - 1$ we have $I_{ar+b-1,1,1}(s) = I_{ar+b-1,2,2}(s) = I_{ar+b-1,2,1}(s) = 0$ and

$$I_{ar+b-1}(s) = I_{ar+b-1,1,2}(s) = (1 - e^{is}) e^{is(a-1)} \Gamma_{1,2}(a)$$

and similarly for $k = ar - b - 1$ we have $I_{ar-b-1,1,1}(s) = I_{ar-b-1,2,2}(s) = I_{ar-b-1,1,2}(s) = 0$ and

$$I_{ar-b-1}(s) = I_{ar-b-1,2,1}(s) = (1 - e^{is}) e^{is(a-1)} \Gamma_{2,1}(a).$$

Then we can write

$$\sum_{k=0}^{\infty} q_k(s) = \sum_{a=1}^{\infty} I_{ar-b-1}(s) + I_{ar-1}(s) + I_{ar+b-1}(s). \quad (2.24)$$

We now have three cases.

Case 1: If $\Gamma_{1,2}(a) = p_2 \alpha^a \gamma_2$ and $\Gamma_{2,1}(a) = p_1 \alpha^{a-1} \gamma_1$ for all $a \geq 1$ ¹¹, then using (2.24) we have

$$\begin{aligned} \sum_{k=0}^{\infty} q_k(s) &= (1 - e^{is}) \sum_{a=1}^{\infty} \left(e^{is(a-1)} \right) (\alpha^a + p_2 \gamma_2 \alpha^a + p_1 \gamma_1 \alpha^{a-1}) \\ &= (1 - e^{is}) (p_1 \gamma_1 + \alpha (p_2 \gamma_2 + 1)) \sum_{a=1}^{\infty} (e^{is} \alpha)^{a-1} \\ &= (p_1 \gamma_1 + \alpha (p_2 \gamma_2 + 1)) \frac{1 - e^{is}}{1 - \alpha e^{is}}. \end{aligned}$$

Case 2: Suppose that $a_{1,2} > 1$ and $\Gamma_{2,1}(a) = p_1 \alpha^{a-1} \gamma_1$ for all $a \geq 1$. Then

$$\Gamma_{1,2}(a) = \begin{cases} p_1 & \text{for all } a < a_{1,2} \\ p_2 \alpha^a \gamma_2 & \text{for all } a \geq a_{1,2}. \end{cases}$$

Then using (2.24) we have

$$\sum_{k=0}^{\infty} q_k(s) = \sum_{a=1}^{a_{1,2}-1} I_{ar+b-1}(s) + \sum_{a=a_{1,2}}^{\infty} I_{ar+b-1}(s) + \sum_{a=1}^{\infty} I_{ar-b-1}(s) + I_{ar-1}(s).$$

The first two sums are given by

$$\begin{aligned} \sum_{a=1}^{a_{1,2}-1} I_{ar+b-1}(s) + \sum_{a=a_{1,2}}^{\infty} I_{ar+b-1}(s) &= \sum_{a=1}^{a_{1,2}-1} (1 - e^{is}) e^{is(a-1)} p_1 + \sum_{a=a_{1,2}}^{\infty} (1 - e^{is}) e^{is(a-1)} p_2 \alpha^a \gamma_2 \\ &= p_1 (1 - e^{is}) \frac{e^{is(a_{1,2}-1)} - 1}{e^{is} - 1} + (1 - e^{is}) p_2 \gamma_2 \alpha \frac{(e^{is} \alpha)^{a_{1,2}-1}}{1 - \alpha e^{is}} \\ &= p_1 \left(1 - e^{is(a_{1,2}-1)} \right) + (1 - e^{is}) p_2 \gamma_2 \alpha \frac{(e^{is} \alpha)^{a_{1,2}-1}}{1 - \alpha e^{is}}, \end{aligned}$$

¹¹ Note that this is the case if $p_1 = p_2$.

and the third of the three sums is given by

$$\begin{aligned}\sum_{a=1}^{\infty} I_{ar-b-1}(s) + I_{ar-1}(s) &= \sum_{a=1}^{\infty} (1 - e^{is}) e^{is(a-1)} (\alpha^a + p_1 \alpha^{a-1} \gamma_1) \\ &= (1 - e^{is}) \left((\alpha + p_1 \gamma_1) \sum_{a=1}^{\infty} (e^{is} \alpha)^{a-1} \right) \\ &= (\alpha + p_1 \gamma_1) \frac{1 - e^{is}}{1 - \alpha e^{is}}.\end{aligned}$$

Combining these sums gives

$$\sum_{k=0}^{\infty} q_k(s) = p_1 \left(1 - e^{is(a_{1,2}-1)} \right) + (1 - e^{is}) p_2 \gamma_2 \alpha \frac{(e^{is} \alpha)^{a_{1,2}-1}}{1 - \alpha e^{is}} + (\alpha + p_1 \gamma_1) \frac{1 - e^{is}}{1 - \alpha e^{is}}.$$

Case 3: Suppose that $\Gamma_{1,2}(a) = p_2 \alpha^a \gamma_2$ for all $a \geq 1$ and $a_{2,1} > 1$. Then

$$\Gamma_{2,1}(a) = \begin{cases} p_2 & \text{for all } a < a_{2,1} \\ p_1 \alpha^{a-1} \gamma_1 & \text{for all } a \geq a_{2,1}. \end{cases}$$

Arguing similarly as in Case 2, we have

$$\sum_{k=0}^{\infty} q_k(s) = p_2 \left(1 - e^{is(a_{2,1}-1)} \right) + (1 - e^{is}) p_1 \gamma_1 \alpha \frac{(e^{is} \alpha)^{a_{2,1}-1}}{1 - \alpha e^{is}} + \alpha (1 + p_2 \gamma_2) \frac{1 - e^{is}}{1 - \alpha e^{is}}.$$

Note that we cannot have a fourth case where both $a_{1,2} > 1$ and $a_{2,1} > 1$. If this were to occur, then we would have that

$$\Gamma_{1,2}(a) = \begin{cases} p_1 & \text{for all } a < a_{1,2} \\ p_2 \alpha^a \gamma_2 & \text{for all } a \geq a_{1,2} \end{cases}$$

and

$$\Gamma_{2,1}(a) = \begin{cases} p_2 & \text{for all } a < a_{2,1} \\ p_1 \alpha^{a-1} \gamma_1 & \text{for all } a \geq a_{2,1}. \end{cases}$$

In particular, the definition of $\Gamma_{1,2}$ and $\Gamma_{2,1}$ for $a = 1 < a_{1,2}, a_{2,1}$ imply that

$$p_1 \leq p_2 \alpha \gamma_2 \quad \text{and} \quad p_2 \leq p_1 \gamma_1.$$

Taken together, and noting that $\alpha, \gamma_1, \gamma_2 \in (0, 1)$, we arrive at the contradiction that

$$p_2 \leq p_1 \gamma_1 \leq p_2 \alpha \gamma_1 \gamma_2.$$

In each of the four cases we then have that the characteristic function $\varphi(s)$ is given by

$$\varphi(s) = \exp \left(-t \left(1 - e^{is} \right) \left(1 - \sum_{k=0}^{\infty} q_k(s) \right) \right),$$

which is clearly not the CF for a standard Poisson or Pólya–Aeppli distribution.

3. The random setting: main results

In this section we now move to the setting of random dynamical systems where we will present our main result. To begin, let $(\Omega, \mathcal{F}, m)$ be a probability space and $\sigma : \Omega \rightarrow \Omega$ an ergodic, invertible map which preserves the measure m , i.e.

$$m \circ \sigma^{-1} = m.$$

For each $\omega \in \Omega$, we take $\mathcal{J}_\omega$ to be a closed subset of a complete metrizable space M such that the map

$$\Omega \ni \omega \longmapsto \mathcal{J}_\omega$$

is a closed random set, i.e. $\mathcal{J}_\omega \subseteq M$ is closed for each $\omega \in \Omega$ and the map $\omega \mapsto \mathcal{J}_\omega$ is measurable (see [20]), and we consider the maps

$$T_\omega : \mathcal{J}_\omega \rightarrow \mathcal{J}_{\sigma\omega}.$$

By $T_\omega^n : \mathcal{J}_\omega \rightarrow \mathcal{J}_{\sigma^n\omega}$ we mean the n -fold composition

$$T_\omega^n := T_{\sigma^n\omega} \circ \cdots \circ T_\omega : \mathcal{J}_\omega \rightarrow \mathcal{J}_{\sigma^n\omega}.$$

Given a set $A \subseteq \mathcal{J}_{\sigma^n\omega}$ we let

$$T_\omega^{-n}(A) := \{x \in \mathcal{J}_\omega : T_\omega^n(x) \in A\}$$

denote the inverse image of A under the map T_ω^n for each $\omega \in \Omega$ and $n \geq 1$. Now let

$$\mathcal{J} := \bigcup_{\omega \in \Omega} \{\omega\} \times \mathcal{J}_\omega \subseteq \Omega \times M,$$

and define the induced skew-product map $T : \mathcal{J} \rightarrow \mathcal{J}$ by

$$T(\omega, x) = (\sigma\omega, T_\omega(x)).$$

Let $\mathcal{B}$ denote the Borel σ -algebra of M and let $\mathcal{F} \otimes \mathcal{B}$ be the product σ -algebra on $\Omega \times M$. Throughout the text we denote Lebesgue measure by Leb . We suppose the following:

(M1) The map $T : \mathcal{J} \rightarrow \mathcal{J}$ is measurable with respect to $\mathcal{F} \otimes \mathcal{B}$.

Definition 3.1. A measure μ on $\Omega \times M$ with respect to the product σ -algebra $\mathcal{F} \otimes \mathcal{B}$ is said to be *random measure* relative to m if it has marginal m , i.e. if

$$\mu \circ \pi_1^{-1} = m.$$

The disintegrations $\{\mu_\omega\}_{\omega \in \Omega}$ of μ with respect to the partition $(\{\omega\} \times M)_{\omega \in \Omega}$ satisfy the following properties:

- (1) For every $B \in \mathcal{B}$, the map $\Omega \ni \omega \mapsto \mu_\omega(B) \in [0, \infty]$ is measurable,
- (2) For m -a.e. $\omega \in \Omega$, the map $\mathcal{B} \ni B \mapsto \mu_\omega(B) \in [0, \infty]$ is a Borel measure.

We say that the random measure $\mu = \{\mu_\omega\}_{\omega \in \Omega}$ is a *random probability measure* if for m -a.e. $\omega \in \Omega$ the fibre measure μ_ω is a probability measure. Given a set $Z = \cup_{\omega \in \Omega} \{\omega\} \times Z_\omega \subseteq \Omega \times M$, we say that the random measure $\mu = \{\mu_\omega\}_{\omega \in \Omega}$ is supported in Z if $\text{supp}(\mu) \subseteq Z$ and consequently $\text{supp}(\mu_\omega) \subseteq Z_\omega$ for m -a.e. $\omega \in \Omega$. We let $\mathcal{P}_\Omega(Z)$ denote the set of all random probability measures supported in Z . We will frequently denote a random measure μ by $\{\mu_\omega\}_{\omega \in \Omega}$.

The following proposition from Crauel [20], shows that a random probability measure $\{\mu_\omega\}_{\omega \in \Omega}$ on $\mathcal{J}$ uniquely identifies a probability measure on $\mathcal{J}$.

Proposition 3.2 ([20], propositions 3.3). *If $\{\mu_\omega\}_{\omega \in \Omega} \in \mathcal{P}_\Omega(\mathcal{J})$ is a random probability measure on $\mathcal{J}$, then for every bounded measurable function $f: \mathcal{J} \rightarrow \mathbb{R}$, the function*

$$\Omega \ni \omega \mapsto \int_{\mathcal{J}_\omega} f(\omega, x) d\mu_\omega(x)$$

is measurable and

$$\mathcal{F} \otimes \mathcal{B} \ni A \mapsto \int_{\Omega} \int_{\mathcal{J}_\omega} \mathbb{1}_A(\omega, x) d\mu_\omega(x) dm(\omega)$$

defines a probability measure on $\mathcal{J}$.

For $f: \mathcal{J} \rightarrow \mathbb{R}$ we let

$$S_n(f_\omega) := \sum_{j=0}^{n-1} f_{\sigma^j(\omega)} \circ T_\omega^j$$

denote the Birkhoff sum of f with respect to T . We will consider a potential of the form $\varphi_0: \mathcal{J} \rightarrow \mathbb{R}$, and for each $n \geq 1$ we consider the weight $g_0^{(n)}: \mathcal{J} \rightarrow \mathbb{R}$ whose disintegrations are given by

$$g_{\omega,0}^{(n)} := \exp(S_n(\varphi_{\omega,0})) = \prod_{j=0}^{n-1} g_{\sigma^j \omega,0}^{(1)} \circ T_\omega^j \quad (3.1)$$

for each $\omega \in \Omega$. We will often denote $g_{\omega,0}^{(1)}$ by $g_{\omega,0}$. We assume there exists a family of Banach spaces $\{\mathcal{B}_\omega, \|\cdot\|_{\mathcal{B}_\omega}\}_{\omega \in \Omega}$ of complex-valued functions on each $\mathcal{J}_\omega$ with $g_{\omega,0} \in \mathcal{B}_\omega$ such that the fiberwise (Perron–Frobenius) transfer operator $\mathcal{L}_{\omega,0}: \mathcal{B}_\omega \rightarrow \mathcal{B}_{\sigma\omega}$ given by

$$\mathcal{L}_{\omega,0}(f)(x) := \sum_{y \in T_\omega^{-1}(x)} f(y) g_{\omega,0}(y), \quad f \in \mathcal{B}_\omega, x \in \mathcal{J}_{\sigma\omega} \quad (3.2)$$

is well defined. Using induction we see that iterates $\mathcal{L}_{\omega,0}^n: \mathcal{B}_\omega \rightarrow \mathcal{B}_{\sigma^n \omega}$ of the transfer operator are given by

$$\mathcal{L}_{\omega,0}^n(f)(x) := \sum_{y \in T_\omega^{-n}(x)} f(y) g_{\omega,0}^{(n)}(y), \quad f \in \mathcal{B}_\omega, x \in \mathcal{J}_{\sigma^n \omega}.$$

We let $\mathcal{B}$ denote the space of functions $f: \mathcal{J} \rightarrow \mathbb{R}$ such that $f_\omega \in \mathcal{B}_\omega$ for each $\omega \in \Omega$ and we define the global transfer operator $\mathcal{L}_0: \mathcal{B} \rightarrow \mathcal{B}$ by

$$(\mathcal{L}_0 f)_\omega(x) := \mathcal{L}_{\sigma^{-1}\omega,0} f_{\sigma^{-1}\omega}(x)$$

for $f \in \mathcal{B}$ and $x \in \mathcal{J}_\omega$. We assume the following measurability assumption:

(M2) For every measurable function $f \in \mathcal{B}$, the map $(\omega, x) \mapsto (\mathcal{L}_0 f)_\omega(x)$ is measurable.

We suppose that the following condition on the existence of a closed conformal measure.

(CCM) There exists a random probability measure $\nu_0 = \{\nu_{\omega,0}\}_{\omega \in \Omega} \in \mathcal{P}_\Omega(\mathcal{J})$ and measurable functions $\lambda_0 : \Omega \rightarrow (0, \infty)$ and $\phi_0 : \mathcal{J} \rightarrow (0, \infty)$ with $\phi_0 \in \mathcal{B}$ such that

$$\mathcal{L}_{\omega,0}(\phi_{\omega,0}) = \lambda_{\omega,0} \phi_{\sigma\omega,0} \quad \text{and} \quad \nu_{\sigma\omega,0}(\mathcal{L}_{\omega,0}(f)) = \lambda_{\omega,0} \nu_{\omega,0}(f)$$

for all $f \in \mathcal{B}_\omega$ where $\phi_{\omega,0}(\cdot) := \phi_0(\omega, \cdot)$. Furthermore, we assume that the fibre measures $\nu_{\omega,0}$ are non-atomic and that $\lambda_{\omega,0} := \nu_{\sigma\omega,0}(\mathcal{L}_{\omega,0}\mathbb{1})$ with $\log \lambda_{\omega,0} \in L^1(m)$. We then define the random probability measure μ_0 on $\mathcal{J}$ by

$$\mu_{\omega,0}(f) := \int_{\mathcal{J}_\omega} f \phi_{\omega,0} d\nu_{\omega,0}, \quad f \in L^1(\nu_{\omega,0}). \quad (3.3)$$

From the definition, one can easily show that μ_0 is T -invariant, that is,

$$\int_{\mathcal{J}_\omega} f \circ T_\omega d\mu_{\omega,0} = \int_{\mathcal{J}_{\sigma\omega}} f d\mu_{\sigma\omega,0}, \quad f \in L^1(\mu_{\sigma\omega,0}). \quad (3.4)$$

Remark 3.3. Our assumption (CCM) has been shown to hold in several random settings: random interval maps [3–5], random subshifts [10, 48], random distance expanding maps [49], random polynomial systems [11], and random transcendental maps [50].

We will also assume that the fiberwise Banach spaces $\mathcal{B}_\omega \subseteq L^\infty(\nu_{\omega,0})$, where $\nu_0 = (\nu_{\omega,0})_{\omega \in \Omega}$ is the random probability measure given by (CCM). We denote the norm on $L^\infty(\nu_{\omega,0})$ by $\|\cdot\|_{\infty,\omega}$.

3.1. Random perturbations

For each $n \in \mathbb{N}$ we let $H_n \subseteq \mathcal{J}$ be measurable with respect to the product σ -algebra $\mathcal{F} \otimes \mathcal{B}$ on $\mathcal{J}$ such that

(A) $H_{n'} \subseteq H_n$ for each $n' \geq n$.

Then the sets $H_{\omega,n} \subseteq \mathcal{J}_\omega$ are uniquely determined by the condition that

$$\{\omega\} \times H_{\omega,n} = H_n \cap (\{\omega\} \times \mathcal{J}_\omega),$$

or equivalently that

$$H_{\omega,n} = \pi_2(H_n \cap (\{\omega\} \times \mathcal{J}_\omega)),$$

where $\pi_2 : \mathcal{J} \rightarrow \mathcal{J}_\omega$ is the projection onto the second coordinate. The sets $H_{\omega,n}$ are then $\nu_{\omega,0}$ -measurable, and (A) implies that

(A') $H_{\omega,n'} \subseteq H_{\omega,n}$ for each $n' \geq n$ and each $\omega \in \Omega$.

For each $\omega \in \Omega$, each $s \in \mathbb{R}$, and each $n \in \mathbb{N}$ we define the perturbed operator $\mathcal{L}_{\omega,n,s} : \mathcal{B}_\omega \rightarrow \mathcal{B}_{\sigma\omega}$

$$\mathcal{L}_{\omega,n,s}(f) := \mathcal{L}_{\omega,0}(f \cdot e^{is\mathbb{1}_{H_{\omega,n}}}).$$

Note that if $s = 0$ then $\mathcal{L}_{\omega,n,0} = \mathcal{L}_{\omega,0}$ for each $n \geq 1$. If we denote

$$S_{\omega,n,k}(x) := \sum_{j=0}^{k-1} \mathbb{1}_{H_{\sigma^j \omega,n}}(T_{\omega}^j x),$$

then we have that

$$\mathcal{L}_{\omega,n,s}^k(f) = \mathcal{L}_{\omega,0}^k(e^{isS_{\omega,n,k}}f). \quad (3.5)$$

Additionally we make the following assumption on our Banach spaces $\mathcal{B}_\omega$: for each $\omega \in \Omega$ we have

- (B) $\mathbb{1} \in \mathcal{B}_\omega$, $\mathbb{1}_{H_{\omega,n}} \in \mathcal{B}_\omega$ for each n , and there exists $C > 0$ such that for m -a.e. $\omega \in \Omega$ we have $\|\mathbb{1}_{H_{\omega,n}}\|_{\mathcal{B}_\omega} \leq C$.

With a view toward applying theorem 2.1.2 in [5] we calculate the following quantities

$$\begin{aligned} \Delta_{\omega,n}(s) &:= \nu_{\sigma\omega,0}((\mathcal{L}_{\omega,0} - \mathcal{L}_{\omega,n,s})(\phi_{\omega,0})) = \nu_{\sigma\omega,0}(\mathcal{L}_{\omega,0}(\phi_{\omega,0}(1 - e^{is\mathbb{1}_{H_{\omega,n}}})) \\ &= \lambda_{\omega,0} \cdot \nu_{\omega,0}(\phi_{\omega,0}(1 - e^{is\mathbb{1}_{H_{\omega,n}}})) = \lambda_{\omega,0} \cdot \mu_{\omega,0}(1 - e^{is\mathbb{1}_{H_{\omega,n}}}) \\ &= \lambda_{\omega,0}(1 - e^{is})\mu_{\omega,0}(H_{\omega,n}), \end{aligned} \quad (3.6)$$

and

$$\begin{aligned} \eta_{\omega,n}(s) &:= \|\nu_{\sigma\omega,0}(\mathcal{L}_{\omega,0} - \mathcal{L}_{\omega,n,s})\|_{\mathcal{B}_\omega^*} = \sup_{\|\psi\|_{\mathcal{B}_\omega} \leq 1} |\nu_{\sigma\omega,0}(\mathcal{L}_{\omega,0}(\psi(1 - e^{is\mathbb{1}_{H_{\omega,n}}}))| \\ &= \lambda_{\omega,0} \cdot \sup_{\|\psi\|_{\mathcal{B}_\omega} \leq 1} |\nu_{\omega,0}(\psi(1 - e^{is\mathbb{1}_{H_{\omega,n}}}))| \\ &= \lambda_{\omega,0} |1 - e^{is}| \sup_{\|\psi\|_{\mathcal{B}_\omega} \leq 1} |\nu_{\omega,0}(\psi \mathbb{1}_{H_{\omega,n}})|, \end{aligned} \quad (3.7)$$

where the final equality follows from the fact that $1 - e^{is\mathbb{1}_{H_{\omega,n}}}(x) = 0$ for $x \notin H_{\omega,n}$.

Remark 3.4. If there exists some measurable, finite m -a.e. function $K : \Omega \rightarrow (0, \infty)$ such that $\|\cdot\|_{\infty,\omega} \leq K_\omega \|\cdot\|_{\mathcal{B}_\omega}$, then the final equality of (3.7) can be bounded by the following:

$$\begin{aligned} \eta_{\omega,n}(s) &\leq \lambda_{\omega,0} |1 - e^{is}| \sup_{\|\psi\|_{\mathcal{B}_\omega} \leq 1} \|\psi\|_{\infty,\omega} \nu_{\omega,0}(H_{\omega,n}) \\ &\leq K_\omega \lambda_{\omega,0} |1 - e^{is}| \nu_{\omega,0}(H_{\omega,n}), \end{aligned} \quad (3.8)$$

Note that the right hand side of (3.8) approaches zero as $n \rightarrow \infty$ if $\nu_{\omega,0}(H_{\omega,n}) \rightarrow 0$ as $n \rightarrow \infty$. Furthermore, if $\mathcal{B}_\omega$ is the space of bounded variation functions, we may take $K_\omega \equiv 2$.

On the other hand, letting $\alpha > 0$ sufficiently small such that $\alpha \|\mathbb{1}\|_{\mathcal{B}_\omega} \leq 1$, in light of (3.7), we see that (for $\psi = \alpha \mathbb{1}$)

$$\lambda_{\omega,0} |1 - e^{is}| \alpha \nu_{\omega,0}(H_{\omega,n}) \leq \eta_{\omega,n}(s). \quad (3.9)$$

In particular, we see that if $\eta_{\omega,n}(s) \rightarrow 0$ as $n \rightarrow \infty$, we must also have that $\nu_{\omega,0}(H_{\omega,n}) \rightarrow 0$.

In what follows, for each $\omega \in \Omega$ and $k \in \mathbb{N}$, we let $\lambda_{\omega,0}^k := \lambda_{\omega,0} \cdots \lambda_{\sigma^{k-1}\omega,0}$. Now for each $\omega \in \Omega$, $n \in \mathbb{N}$, $s \in \mathbb{R}$, and $k \in \mathbb{N}$, we define the quantities $q_{\omega,n}^{(k)}(s)$ by the following formula:

$$\begin{aligned} q_{\omega,n}^{(k)}(s) &:= \frac{\nu_{\sigma\omega,0} \left((\mathcal{L}_{\omega,0} - \mathcal{L}_{\omega,n,s}) \left(\mathcal{L}_{\sigma^{-k}\omega,n,s}^k \right) \left(\mathcal{L}_{\sigma^{-(k+1)}\omega,0} - \mathcal{L}_{\sigma^{-(k+1)}\omega,n,s} \right) \left(\phi_{\sigma^{-(k+1)}\omega,0} \right) \right)}{\Delta_{\omega,n}(s)} \\ &= \frac{\lambda_{\sigma^{-(k+1)}\omega,0}^{k+2} \cdot \mu_{\sigma^{-(k+1)}\omega,0} \left(\left(1 - e^{\text{is} \mathbb{I}_{H_{\omega,n}} \circ T_{\sigma^{-(k+1)}\omega}^{k+1}} \right) \left(e^{\text{is} S_{\sigma^{-k}\omega,n,k} \circ T_{\sigma^{-(k+1)}\omega}} \right) \left(1 - e^{\text{is} \mathbb{I}_{H_{\sigma^{-(k+1)}\omega,n}} \right) \right)}{\lambda_{\omega,0} (1 - e^{\text{is}}) \mu_{\omega,0}(H_{\omega,n})} \\ &= \frac{\lambda_{\sigma^{-(k+1)}\omega,0}^{k+1} \cdot \mu_{\sigma^{-(k+1)}\omega,0} \left(\left(1 - e^{\text{is} \mathbb{I}_{H_{\omega,n}} \circ T_{\sigma^{-(k+1)}\omega}^{k+1}} \right) \left(e^{\text{is} S_{\sigma^{-k}\omega,n,k} \circ T_{\sigma^{-(k+1)}\omega}} \right) \left(1 - e^{\text{is} \mathbb{I}_{H_{\sigma^{-(k+1)}\omega,n}} \right) \right)}{(1 - e^{\text{is}}) \mu_{\omega,0}(H_{\omega,n})} \\ &= \frac{\lambda_{\sigma^{-(k+1)}\omega,0}^{k+1} (1 - e^{\text{is}})^2 \mu_{\sigma^{-(k+1)}\omega,0} \left(\left(\mathbb{I}_{H_{\omega,n}} \circ T_{\sigma^{-(k+1)}\omega}^{k+1} \right) \left(e^{\text{is} S_{\sigma^{-k}\omega,n,k} \circ T_{\sigma^{-(k+1)}\omega}} \right) \left(\mathbb{I}_{H_{\sigma^{-(k+1)}\omega,n}} \right) \right)}{(1 - e^{\text{is}}) \mu_{\omega,0}(H_{\omega,n})} \\ &= \frac{\lambda_{\sigma^{-(k+1)}\omega,0}^{k+1} (1 - e^{\text{is}}) \mu_{\sigma^{-(k+1)}\omega,0} \left(\left(\mathbb{I}_{T_{\sigma^{-(k+1)}\omega}^{-(k+1)}(H_{\omega,n})} \right) \left(e^{\text{is} S_{\sigma^{-k}\omega,n,k} \circ T_{\sigma^{-(k+1)}\omega}} \right) \left(\mathbb{I}_{H_{\sigma^{-(k+1)}\omega,n}} \right) \right)}{\mu_{\omega,0}(H_{\omega,n})}. \end{aligned}$$

For notational convenience we define the quantity $\hat{q}_{\omega,n}^{(k)}(s)$ by

$$\hat{q}_{\omega,n}^{(k)}(s) := \frac{(1 - e^{\text{is}}) \mu_{\sigma^{-(k+1)}\omega,0} \left(\left(\mathbb{I}_{T_{\sigma^{-(k+1)}\omega}^{-(k+1)}(H_{\omega,n})} \right) \left(e^{\text{is} S_{\sigma^{-k}\omega,n,k} \circ T_{\sigma^{-(k+1)}\omega}} \right) \left(\mathbb{I}_{H_{\sigma^{-(k+1)}\omega,n}} \right) \right)}{\mu_{\omega,0}(H_{\omega,n})}, \quad (3.10)$$

and thus we have that

$$\hat{q}_{\omega,n}^{(k)}(s) = \left(\lambda_{\sigma^{-(k+1)}\omega,0}^{k+1} \right)^{-1} q_{\omega,n}^{(k)}(s).$$

We assume the following conditions hold for each $n \in \mathbb{N}$ sufficiently large and each $s \in \mathbb{R}$:

(C1) There exists $C_1 \geq 1$ such that for m -a.e. $\omega \in \Omega$ we have

$$C_1^{-1} \leq \mathcal{L}_{\omega,0} \mathbb{1} \leq C_1.$$

(C2) There exist $\nu_{\omega,n,s} \in \mathcal{B}_{\omega}^*$ (the dual of $\mathcal{B}_{\omega}$) and measurable functions $\lambda_{n,s} : \Omega \rightarrow \mathbb{C} \setminus \{0\}$ with $\log |\lambda_{\omega,n,s}| \in L^1(m)$ and $\phi_{n,s} : \mathcal{T} \rightarrow \mathbb{C}$ such that

$$\mathcal{L}_{\omega,n,s}(\phi_{\omega,n,s}) = \lambda_{\omega,n,s} \phi_{\sigma\omega,n,s} \quad \text{and} \quad \nu_{\sigma\omega,n,s}(\mathcal{L}_{\omega,n,s}(f)) = \lambda_{\omega,n,s} \nu_{\omega,n,s}(f)$$

for all $f \in \mathcal{B}_{\omega}$. Furthermore we assume that for m -a.e. $\omega \in \Omega$

$$\nu_{\omega,0}(\phi_{\omega,n,s}) = 1 \quad \text{and} \quad \nu_{\omega,0}(\mathbb{1}) = 1.$$

(C3) There exist operators $\mathcal{Q}_{\omega,0}, \mathcal{Q}_{\omega,n,s} : \mathcal{B}_{\omega} \rightarrow \mathcal{B}_{\sigma\omega}$ such that for m -a.e. $\omega \in \Omega$ and each $f \in \mathcal{B}_{\omega}$ we have

$$\lambda_{\omega,0}^{-1} \mathcal{L}_{\omega,0}(f) = \nu_{\omega,0}(f) \cdot \phi_{\sigma\omega,0} + \mathcal{Q}_{\omega,0}(f)$$

and

$$\lambda_{\omega,n,s}^{-1} \mathcal{L}_{\omega,n,s}(f) = \nu_{\omega,n,s}(f) \cdot \phi_{\sigma\omega,n,s} + \mathcal{Q}_{\omega,n,s}(f).$$

Furthermore, for m -a.e. $\omega \in \Omega$ we have

$$Q_{\omega,0}(\phi_{\omega,0}) = 0 \quad \text{and} \quad \nu_{\sigma\omega,0}(Q_{\omega,0}(f)) = 0$$

and

$$Q_{\omega,n,s}(\phi_{\omega,n,s}) = 0 \quad \text{and} \quad \nu_{\sigma\omega,n,s}(Q_{\omega,n,s}(f)) = 0.$$

(C4) There exists $C > 0$ and a summable sequence $\alpha(N) > 0$ (independent of ω) with $\alpha := \sum_{N=1}^{\infty} \alpha(N) < \infty$ such that for each $f \in \mathcal{B}$, for m -a.e. $\omega \in \Omega$, and each $N \in \mathbb{N}$

$$\|Q_{\omega,0}^N f_{\omega}\|_{\infty, \sigma^N \omega} \leq C \alpha(N) \|f_{\omega}\|_{\mathcal{B}_{\omega}}.$$

and

$$\sup_{n \geq 0} \|Q_{\omega,n,s}^N f_{\omega}\|_{\infty, \sigma^N \omega} \leq C \alpha(N) \|f_{\omega}\|_{\mathcal{B}_{\omega}}.$$

(C5) There exists $C_2 \geq 1$ such that

$$\|\phi_{\omega,0}\|_{\mathcal{B}_{\omega}} \leq C_2 \quad \text{and} \quad \sup_{n \geq 0} \|\phi_{\omega,n,s}\|_{\infty, \omega} \leq C_2$$

for m -a.e. $\omega \in \Omega$.

(C6) For m -a.e. $\omega \in \Omega$ we have

$$\lim_{n \rightarrow \infty} \eta_{\omega,n}(s) = 0.$$

(C7) There exists $C_3 \geq 1$ such that

$$\operatorname{ess\,inf}_{\omega} \phi_{\omega,0} \geq C_3^{-1} > 0.$$

(C8) For m -a.e. $\omega \in \Omega$ we have that the limit $\hat{q}_{\omega,0}^{(k)}(s) := \lim_{n \rightarrow 0} \hat{q}_{\omega,n}^{(k)}(s)$ exists for each $k \geq 0$, where $\hat{q}_{\omega,n}^{(k)}(s)$ is as in (3.10).

(S) For any fixed random scaling function $t \in L^{\infty}(m)$ with $t > 0$, we may find a sequence of functions $\xi_n \in L^{\infty}(m)$ and a constant $W < \infty$ satisfying

$$\mu_{\omega,0}(H_{\omega,n}) = (t_{\omega} + \xi_{\omega,n})/n, \text{ for a.e. } \omega \text{ and each } n \geq 1,$$

where:

- (i) $\lim_{n \rightarrow \infty} \xi_{\omega,n} = 0$ for a.e. ω and
- (ii) $|\xi_{\omega,n}| \leq W$ for a.e. ω and all $n \geq 1$.

Remark 3.5. Assumption (C4) can easily be shown to imply the following decay of correlations condition:

- For m -a.e. every $N \in \mathbb{N}$, every $f \in L^1(\mu)$ and every $h \in \mathcal{B}$ we have

$$\begin{aligned} & \left| \mu_{\omega,0} \left(\left(f_{\sigma^N(\omega)} \circ T_{\omega}^N \right) h_{\omega} \right) - \mu_{\sigma^N(\omega),0} \left(f_{\sigma^N(\omega)} \right) \mu_{\omega,0} (h_{\omega}) \right| \\ & \leq C \|f_{\sigma^N \omega}\|_{L^1(\mu_{\sigma^N \omega,0})} \|h_{\omega}\|_{\mathcal{B}_{\omega}} \alpha(N). \end{aligned} \quad (3.11)$$

Remark 3.6. (*On the scaling (S) for non-random t*) If t is non-random, there may be situations in which it is easier to check (S)(i) using natural annealed quantities. A scaling that has been proposed in the quenched random setting reads (in our notation) for $n \geq 1$:

$$\lim_{n \rightarrow \infty} n \int \mu_{\omega,0}(H_{\omega,n}) \, dm(\omega) = t \quad (3.12)$$

see [2, 55].

We claim that the scaling (3.12) together with an additional *annealed* assumption: for any $t > 0$, there exist $\kappa(t) > 0$ and a bounded constant $C(t)$ such that

$$\int |n\mu_{\omega,0}(H_{\omega,n}) - t| dm(\omega) \leq \frac{C(t)}{n^{1+\kappa(t)}}, \quad (3.13)$$

imply condition (S)(i). It is sufficient to show, by Borel–Cantelli, that for any $\varepsilon > 0$, $\sum_n m(|n\mu_{\omega,0}(H_{\omega,n}) - t| > \varepsilon) < \infty$. By Tchebychev's inequality we have

$$m(|n\mu_{\omega,0}(H_{\omega,n}) - t| > \varepsilon) \leq \frac{1}{\varepsilon} \mathbb{E}(|n\mu_{\omega,0}(H_{\omega,n}) - t|).$$

Thus (3.12) and (3.13) imply part (i) of our condition (S):

$$n\mu_{\omega,0}(H_{\omega,n}) \rightarrow t, \text{ a.s.}$$

To ensure that part (ii) of (S) holds, it suffices to assume that there exists $K > 0$ such that

$$\limsup_{n \rightarrow \infty} n\mu_{\omega,0}(H_{\omega,n}) \leq K + t.$$

Indeed, this immediately implies that

$$-K \leq n\mu_{\omega,0}(H_{\omega,n}) - t \leq K$$

for all n sufficiently large.

In the other direction, it is immediate to see that condition (S) (with non-random t) implies a limit version of (3.12), namely $n \int \mu_{\omega,0}(H_{\omega,n}) dm(\omega) \rightarrow t$, as $n \rightarrow \infty$.

Definition 3.7. We will call the collection $(\Omega, m, \sigma, \mathcal{J}, T, \mathcal{B}, \mathcal{L}_0, \nu_0, \phi_0, H_n)$ a random perturbed system if the assumptions (M1), (M2), (CCM), (A), (B), (C1)–(C7) are satisfied.

Remark 3.8. See [5] for examples of systems which satisfy the assumptions of definition 3.7 for perturbations similar to what we consider here.

Note that

$$\begin{aligned} \lambda_{\omega,0} - \lambda_{\omega,n,s} &= \lambda_{\omega,0} \nu_{\omega,0}(\phi_{\omega,n,s}) - \nu_{\sigma\omega,0}(\lambda_{\omega,n,s} \phi_{\sigma\omega,n,s}) \\ &= \nu_{\sigma\omega,0}(\mathcal{L}_{\omega,0}(\phi_{\omega,n,s})) - \nu_{\sigma\omega,0}(\mathcal{L}_{\omega,n,s}(\phi_{\omega,n,s})) \\ &= \nu_{\sigma\omega,0}((\mathcal{L}_{\omega,0} - \mathcal{L}_{\omega,n,s})(\phi_{\omega,n,s})). \end{aligned} \quad (3.14)$$

It then follows from (3.7), (3.8), and (3.14) that

$$|\lambda_{\omega,0} - \lambda_{\omega,n,s}| \leq C_2 \eta_{\omega,n}(s). \quad (3.15)$$

Taking (3.15) together with the assumption (C6), we see that for each $s \in \mathbb{R}$

$$\lambda_{\omega,n,s} \rightarrow \lambda_{\omega,0} \quad (3.16)$$

as $n \rightarrow \infty$. Using (CCM), (3.5), (C3), and (C2), for $\psi_\omega \in \mathcal{B}_\omega$ we can write

$$\begin{aligned} \nu_{\omega,0}(\psi_\omega e^{isS_{\omega,n,k}}) &= (\lambda_{\omega,0}^k)^{-1} \nu_{\sigma^k \omega,0}(\mathcal{L}_{\omega,0}^k(\psi_\omega e^{isS_{\omega,n,k}})) = (\lambda_{\omega,0}^k)^{-1} \nu_{\sigma^k \omega,0}(\mathcal{L}_{\omega,n,s}^k \psi_\omega) \\ &= \frac{\lambda_{\omega,n,s}^k}{\lambda_{\omega,0}^k} (\nu_{\omega,n,s}(\psi_\omega) \nu_{\sigma^k \omega,0}(\phi_{\sigma^k \omega,n,s}) + \nu_{\sigma^k \omega,0}(\mathcal{Q}_{\omega,n,s}^k \psi_\omega)) \\ &= \frac{\lambda_{\omega,n,s}^k}{\lambda_{\omega,0}^k} (\nu_{\omega,n,s}(\psi_\omega) + \nu_{\sigma^k \omega,0}(\mathcal{Q}_{\omega,n,s}^k \psi_\omega)). \end{aligned}$$

In particular, if $\psi_\omega = \mathbb{1}$ we have

$$\nu_{\omega,0}(e^{isS_{\omega,n,k}}) = \frac{\lambda_{\omega,n,s}^k}{\lambda_{\omega,0}^k} (\nu_{\omega,n,s}(\mathbb{1}) + \nu_{\sigma^k \omega,0}(\mathcal{Q}_{\omega,n,s}^k \mathbb{1})), \quad (3.17)$$

and if $\psi_\omega = \phi_{\omega,0}$ we have

$$\mu_{\omega,0}(e^{isS_{\omega,n,k}}) = \frac{\lambda_{\omega,n,s}^k}{\lambda_{\omega,0}^k} (\nu_{\omega,n,s}(\phi_{\omega,0}) + \nu_{\sigma^k \omega,0}(\mathcal{Q}_{\omega,n,s}^k \phi_{\omega,0})). \quad (3.18)$$

In order to apply theorem 2.1.2 in [5], we need to check assumptions (P1)–(P9), however as it is clear that our assumptions (C1)–(C8) directly imply (P1)–(P6) as well as (P9), we have only to check the assumptions (P7) and (P8) of [5]. This is done in the following lemma.

Lemma 3.9. *Given a random perturbed system $(\Omega, m, \sigma, \mathcal{I}, T, \mathcal{B}, \mathcal{L}_0, \nu_0, \phi_0, H_n)$, for m -a.e. $\omega \in \Omega$ and each $s \in \mathbb{R}$ we have that*

$$\lim_{n \rightarrow \infty} \nu_{\omega,n,s}(\phi_{\omega,0}) = 1, \quad (3.19)$$

and

$$\lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} (\Delta_{\omega,n}(s))^{-1} \nu_{\sigma \omega,0} \left((\mathcal{L}_{\omega,0} - \mathcal{L}_{\omega,n,s}) \left(\mathcal{Q}_{\sigma^{-k} \omega,n,s}^k \phi_{\sigma^{-k} \omega,0} \right) \right) = 0. \quad (3.20)$$

Thus, (P7) and (P8) of [5] follow from (3.19) and (3.20) respectively.

Proof. First, using (3.6), we note that if $\mu_{\omega,0}(H_{\omega,n}) > 0$ then $\Delta_{\omega,n}(s) \neq 0$. To prove (3.19), we note that for fixed $k \in \mathbb{N}$ we have

$$\lim_{n \rightarrow \infty} \frac{\lambda_{\omega,0}^k}{\lambda_{\omega,n,s}^k} \mu_{\omega,0}(e^{isS_{\omega,n,k}}) = 1 \quad (3.21)$$

since $\lambda_{\omega,0}^k / \lambda_{\omega,n,s}^k \rightarrow 1$ (by (3.16)) and since (3.9) and (C6) together imply that, again noting that k is fixed, $\mu_{\omega,0}(e^{isS_{\omega,n,k}}) \rightarrow 1$ as $n \rightarrow \infty$. Using (3.18) we can write

$$\nu_{\omega,n,s}(\phi_{\omega,0}) = \frac{\lambda_{\omega,0}^k}{\lambda_{\omega,n,s}^k} \mu_{\omega,0}(e^{isS_{\omega,n,k}}) - \nu_{\sigma^k \omega,0}(\mathcal{Q}_{\omega,n,s}^k(\phi_{\omega,0})), \quad (3.22)$$

and thus using (3.21) and (C4), for each $\omega \in \Omega$ and each $k \in \mathbb{N}$ we can write

$$\begin{aligned} \lim_{n \rightarrow \infty} |1 - \nu_{\omega,n,s}(\phi_{\omega,0})| &\leq \lim_{n \rightarrow \infty} \left| 1 - \frac{\lambda_{\omega,0}^k}{\lambda_{\omega,n,s}^k} \mu_{\omega,0}(\mathbf{e}^{isS_{\omega,n,k}}) \right| + \|\mathcal{Q}_{\omega,n,s}^k(\phi_{\omega,0})\|_{\infty, \sigma^k \omega} \\ &\leq C_{\phi_0} \alpha(k) \|\phi_{\omega,0}\|_{\mathcal{B}_\omega}. \end{aligned}$$

As this holds for each $k \in \mathbb{N}$ and as the right-hand side of the previous equation goes to zero as $k \rightarrow \infty$, we must in fact have that

$$\lim_{n \rightarrow \infty} |1 - \nu_{\omega,n,s}(\phi_{\omega,0})| = 0,$$

which yields the first claim.

Now, for the second claim, using (3.6), we note that (C7) implies

$$\begin{aligned} &\left| (\Delta_{\omega,n}(s))^{-1} \nu_{\sigma\omega,0} \left((\mathcal{L}_{\omega,0} - \mathcal{L}_{\omega,n,s}) \left(\mathcal{Q}_{\sigma^{-N}\omega,\varepsilon}^k \phi_{\sigma^{-k}\omega,0} \right) \right) \right| \\ &= \left| \frac{\nu_{\sigma\omega,0} \left(\mathcal{L}_{\omega,0} \left((1 - \mathbf{e}^{is\mathbb{I}_{H_{\omega,n}}}) \mathcal{Q}_{\sigma^{-k}\omega,n,s}^k (\phi_{\sigma^{-k}\omega,0}) \right) \right)}{\lambda_{\omega,0} (1 - \mathbf{e}^{is}) \mu_{\omega,0}(H_{\omega,n})} \right| \\ &= \left| \frac{\nu_{\omega,0} \left((1 - \mathbf{e}^{is\mathbb{I}_{H_{\omega,n}}}) \mathcal{Q}_{\sigma^{-k}\omega,n,s}^k (\phi_{\sigma^{-k}\omega,0}) \right)}{(1 - \mathbf{e}^{is}) \mu_{\omega,0}(H_{\omega,n})} \right| \\ &\leq \frac{\nu_{\omega,0}(H_{\omega,n})}{\mu_{\omega,0}(H_{\omega,n})} \|\mathcal{Q}_{\sigma^{-k}\omega,n,s}^k (\phi_{\sigma^{-k}\omega,0})\|_{\infty, \omega} \leq C_3 \|\mathcal{Q}_{\sigma^{-k}\omega,n,s}^k (\phi_{\sigma^{-k}\omega,0})\|_{\infty, \omega}. \end{aligned}$$

Thus, letting $n \rightarrow \infty$ first and then $k \rightarrow \infty$, the second claim follows from (C4). $\square$

Now, for all $\ell \geq 0$, we define the following:

$$\beta_{\omega,n}^{(k)}(\ell) := \frac{\mu_{\sigma^{-(k+1)}\omega,0} \left(\left\{ x \in H_{\sigma^{-(k+1)}\omega,n} : T_{\sigma^{-(k+1)}\omega}^{k+1}(x) \in H_{\omega,n}, \sum_{j=1}^k \mathbb{1}_{H_{\sigma^{-(k+1)}\omega,n}}(T_{\sigma^{-(k+1)}\omega}^j(x)) = \ell \right\} \right)}{\mu_{\omega,0}(H_{\omega,n})}.$$

Thus, in light of (3.10), we have that

$$\hat{q}_{\omega,n}^{(k)}(s) = (1 - \mathbf{e}^{is}) \sum_{\ell=0}^k \mathbf{e}^{i\ell s} \beta_{\omega,n}^{(k)}(\ell). \quad (3.23)$$

Note that

$$\sum_{\ell=0}^k \beta_{\omega,n}^{(k)}(\ell) = \frac{\mu_{\sigma^{-(k+1)}\omega,0} \left(H_{\sigma^{-(k+1)}\omega,n} \cap T_{\sigma^{-(k+1)}\omega}^{-(k+1)}(H_{\omega,n}) \right)}{\mu_{\omega,0}(H_{\omega,n})}.$$

Using the decay of correlations condition (3.11), which follows from our assumption (C4), together with the assumption (B) and the T -invariance of μ_0 , gives that

$$\begin{aligned}
& \sum_{\ell=0}^k \beta_{\omega,n}^{(k)}(\ell) \\
& \leq \frac{\mu_{\omega,0}(H_{\omega,n}) \mu_{\sigma^{-(k+1)}\omega,0}(H_{\sigma^{-(k+1)}\omega,n}) + C \mu_{\omega,0}(H_{\omega,n}) \|\mathbb{1}_{H_{\sigma^{-(k+1)}\omega,n}}\|_{\mathcal{B}_{\sigma^{-(k+1)}\omega}} \alpha(k+1)}{\mu_{\omega,0}(H_{\omega,n})} \\
& = \mu_{\sigma^{-(k+1)}\omega,0}(H_{\sigma^{-(k+1)}\omega,n}) + C \|\mathbb{1}_{H_{\sigma^{-(k+1)}\omega,n}}\|_{\mathcal{B}_{\sigma^{-(k+1)}\omega}} \alpha(k+1) \\
& \leq \mu_{\sigma^{-(k+1)}\omega,0}(H_{\sigma^{-(k+1)}\omega,n}) + C^2 \alpha(k+1). \tag{3.24}
\end{aligned}$$

In view of (3.23), we now claim that for each $0 \leq \ell \leq k$ the quantities $\beta_{\omega,n}^{(k)}(\ell)$ converge as $n \rightarrow \infty$.

Lemma 3.10. *If (C8) holds, then the limit*

$$\beta_{\omega,0}^{(k)}(\ell) := \lim_{n \rightarrow \infty} \beta_{\omega,n}^{(k)}(\ell)$$

exists for each $0 \leq \ell \leq k$.

Proof. Note that (3.23) applied to $s = 1, 2, \dots, k+1$, yields, for each n, k, ω ,

$$\begin{pmatrix} \hat{q}_{\omega,n}^{(k)}(1) \\ \hat{q}_{\omega,n}^{(k)}(2) \\ \vdots \\ \hat{q}_{\omega,n}^{(k)}(k+1) \end{pmatrix} = D_k M_k \begin{pmatrix} \beta_{\omega,n}^{(k)}(0) \\ \beta_{\omega,n}^{(k)}(1) \\ \vdots \\ \beta_{\omega,n}^{(k)}(k) \end{pmatrix}, \tag{3.25}$$

where

$$D_k = \begin{pmatrix} 1 - e^i & 0 & \dots & 0 \\ 0 & 1 - e^{2i} & \dots & 0 \\ \vdots & & & \\ 0 & 0 & \dots & 1 - e^{(k+1)i} \end{pmatrix} \quad \text{and} \quad M_k = \begin{pmatrix} 1 & e^i & \dots & e^{ki} \\ 1 & e^{2i} & \dots & e^{2ki} \\ \vdots & & & \\ 1 & e^{(k+1)i} & \dots & e^{(k+1)ki} \end{pmatrix}.$$

Clearly, D_k is invertible. The next paragraph shows that M_k is invertible. Thus, pre-multiplying (3.25) by $M_k^{-1}D_k^{-1}$ yields an expression for the β_n 's as linear combinations of $\hat{q}_n$'s. Taking limits as $n \rightarrow \infty$, and recalling that (C8) ensures the limits exist for $\hat{q}_n$'s, the lemma follows.

To finish the proof, we show that M_k is invertible. This is a consequence of (Baker's version of) Lindemann–Weierstrass theorem [9, theorem 1.4], stating that if $a_1, \dots, a_N$ are algebraic numbers, and $\alpha_1, \dots, \alpha_N$ are distinct algebraic numbers, then the equation

$$a_1 e^{\alpha_1} + a_2 e^{\alpha_2} + \dots + a_N e^{\alpha_N} = 0$$

has only the trivial solution $a_i = 0$ for all $1 \leq i \leq N$. Indeed, $\det M_k = \sum_{j=0}^N a_j e^{ij}$, where $N = \sum_{\ell=0}^k \ell(\ell+1)$, a_j is an integer for each $0 \leq j \leq N$, and $a_N = 1$. Thus, the Lindemann–Weierstrass theorem (with $\alpha_j = ij$) implies $\det M_k \neq 0$ and thus M_k is invertible. $\square$

It follows from lemma 3.10 that (assuming (C8) holds) we have

$$\hat{q}_{\omega,0}^{(k)}(s) = \lim_{n \rightarrow \infty} \hat{q}_{\omega,n}^{(k)}(s) = (1 - e^{is}) \sum_{\ell=0}^k e^{i\ell s} \beta_{\omega,0}^{(k)}(\ell),$$

and furthermore, it follows from (3.24) that

$$\sum_{\ell=0}^k \beta_{\omega,0}^{(k)}(\ell) = \lim_{n \rightarrow \infty} \sum_{\ell=0}^k \beta_{\omega,n}^{(k)}(\ell) \leq C^2 \alpha(k+1). \quad (3.26)$$

Denote

$$\Sigma_{\omega} := \sum_{k=0}^{\infty} \sum_{\ell=0}^k \beta_{\omega,0}^{(k)}(\ell). \quad (3.27)$$

Hence, by the assumption of the summability of the $\alpha(k)$ coming from (C4), we have Σ_{ω} is well defined and

$$0 \leq \Sigma_{\omega} \leq \sum_{k=0}^{\infty} C^2 \alpha(k+1) = C^2 \alpha < \infty. \quad (3.28)$$

Thus we have that

$$\theta_{\omega}(s) := 1 - \sum_{k=0}^{\infty} \hat{q}_{\omega,0}^{(k)}(s) = 1 - (1 - e^{is}) \sum_{k=0}^{\infty} \sum_{\ell=0}^k e^{i\ell s} \beta_{\omega,0}^{(k)}(\ell), \quad (3.29)$$

and furthermore we have shown that $|\theta_{\omega}(s)|$ is bounded uniformly in ω and s .

Remark 3.11. Note that the $\beta_{\omega,0}^{(k)}(0)$ are the $\hat{q}_{\omega,0}^{(k)}$ from [5], and thus the standard extremal index $\theta_{\omega,0}$ from [5] is given by

$$\theta_{\omega,0} := 1 - \sum_{k=0}^{\infty} \beta_{\omega,0}^{(k)}(0). \quad (3.30)$$

Remark 3.12. From the calculation of (3.24) and (3.28), we note that the assumption (B) can be weakened from the uniform requirement that $\|\mathbb{1}_{H_{\omega,n}}\|_{\mathcal{B}_{\omega}} \leq C$. All that is required is that

$$\sum_{k=0}^{\infty} \|\mathbb{1}_{H_{\sigma^{-(k+1)}\omega,n}}\|_{\mathcal{B}_{\sigma^{-(k+1)}\omega}} \alpha(k+1) < \infty$$

for each $n \in \mathbb{N}$ and m -a.e. $\omega \in \Omega$.

Under our assumptions (C1)–(C8) we are able to apply theorem 2.1.2 in [5] to get the following result.

Theorem 3.13. Suppose that $(\Omega, m, \sigma, \mathcal{J}, T, \mathcal{B}, \mathcal{L}_0, \nu_0, \phi_0, H_n)$ is a random perturbed system. If (C8) holds, then for each $s \in \mathbb{R}$ and m -a.e. $\omega \in \Omega$

$$\lim_{n \rightarrow \infty} \frac{\lambda_{\omega,0} - \lambda_{\omega,n,s}}{\lambda_{\omega,0} \mu_{\omega,0}(H_{\omega,n})} = (1 - e^{is}) \theta_{\omega}(s).$$

Now following the proof of [5, theorem 2.4.5] and making the necessary changes to suit the perturbations of our current setting, assuming (C1)–(C8) and (S), we obtain the following.

Theorem 3.14. *Given the random perturbed system $(\Omega, m, \sigma, \mathcal{J}, T, \mathcal{B}, \mathcal{L}_0, \nu_0, \phi_0, H_n)$ that also satisfies assumptions (C8) and (S), we have that for each $s \in \mathbb{R}$ and m -a.e. $\omega \in \Omega$*

$$\begin{aligned} \lim_{n \rightarrow \infty} \nu_{\omega,0} (e^{isS_{\omega,n,n}}) &= \lim_{n \rightarrow \infty} \mu_{\omega,0} (e^{isS_{\omega,n,n}}) = \lim_{n \rightarrow \infty} \frac{\lambda_{\omega,n,s}^n}{\lambda_{\omega,0}^n} \\ &= \exp \left(- (1 - e^{is}) \int_{\Omega} t_{\omega} \theta_{\omega}(s) \, dm(\omega) \right). \end{aligned} \quad (3.31)$$

Denote the right hand side of (3.31) by

$$\varphi(s) := \begin{cases} 1 & \text{if } s = 0, \\ \exp \left(- (1 - e^{is}) \int_{\Omega} t_{\omega} \theta_{\omega}(s) \, dm(\omega) \right) & \text{if } s \neq 0. \end{cases} \quad (3.32)$$

Note that since $|\theta_{\omega}(s)|$ is bounded above and below uniformly in ω and s , and since $t \in L^{\infty}(m)$, we have that $|\int_{\Omega} t_{\omega} \theta_{\omega}(s) \, dm(\omega)| < \infty$, which further implies that $\varphi(s)$ is continuous at $s = 0$.

From theorem 3.14, we get that $\varphi(s)$ (which does not depend on ω) is the limit of the characteristic functions of the (ω -dependent) random variables

$$Z_{\omega,n} := S_{\omega,n,n}(x) = \sum_{j=0}^{n-1} \mathbb{1}_{H_{\sigma^j \omega,n}}(T_{\omega}^j x).$$

Since $\varphi(s)$ is continuous at $s = 0$, it follows from the Lévy Continuity Theorem (see [47, theorem 3.6.1]) that φ is the characteristic function of an ω -independent random variable Z on some probability space $(\Gamma', \mathcal{B}', \mathbb{P}')$ to which the sequence of random variables $(Z_{\omega,n})_{n=1}^{\infty}$ (for m -a.e. $\omega \in \Omega$) converge in distribution.

We will denote the distribution of Z by ν_Z . The random variable Z is non-negative and integer valued as it is the distributional limit of a sequence of integer-valued random variables. Furthermore, for m -a.e. $\omega \in \Omega$ we have that

$$\mathbb{P}'(Z = k) = \nu_Z(\{k\}) = \lim_{n \rightarrow \infty} \mu_{\omega,0}(Z_{\omega,n} = k).$$

If no confusion arises, we will denote the underlying probability with $\mathbb{P}$, instead of $\mathbb{P}'$, and its moments with $\mathbb{E}$, Var , etc which are actually computed with the distribution ν_Z .

Now by [24, section 12.2] (see also [57, theorem 3.1]) we see that Z is clearly infinitely divisible, i.e. for each $N \in \mathbb{N}$ we have that $\varphi(s) = (\varphi_N(s))^N$, where φ_N is the CF¹² given by

$$\varphi_N(s) := \left(\exp \left(- (1 - e^{is}) \int_{\Omega} \frac{t_{\omega}}{N} \theta_{\omega}(s) \, dm(\omega) \right) \right),$$

and thus we have that Z is a compound Poisson random variable.

Remark 3.15. Using the Lévy inversion formula we can calculate the probability mass function associated to the random variable Z , that is for each $k \in \{0, 1, 2, \dots\}$ we have

$$\mathbb{P}(Z = k) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T e^{-isk} \exp \left(- (1 - e^{is}) \int_{\Omega} t_{\omega} \theta_{\omega}(s) \, dm(\omega) \right) ds.$$

¹² Note that since $\varphi_N(s) = \lim_{n \rightarrow \infty} \mu_{\omega,0}(e^{is \frac{1}{N} S_{\omega,n,n}})$, one can argue that φ_N is a CF in the same manner that was done for φ .

Remark 3.16. In view of (3.32) we see that the map $s \mapsto \varphi(s)$ is N -times differentiable at $s = 0$ if the map $s \mapsto \theta_\omega(s)$ is N -times differentiable at $s = 0$. It is well known that if N is even, then the RV Z has finite moments up to order N , and if N is odd, then Z has finite moments up to order $N - 1$ (see, for example, [47, theorem 2.3.3]). Since the differentiability of $\theta_\omega(s)$ implies the differentiability of φ , in this case we have that $D_s^k \varphi(0) = i^k \mathbb{E}(Z^k)$ for $k \leq N$, where D_s denotes the derivative operator with respect to s . In light of (3.29) and the fact that $D_s^N e^{i\ell s} = (i\ell)^N e^{i\ell s}$, it is clear that the map $s \mapsto \theta_\omega(s)$ is N -times differentiable at $s = 0$ if

$$\sum_{k=0}^{\infty} \sum_{\ell=0}^k \ell^N \beta_{\omega,0}^{(k)}(\ell) < \infty. \quad (3.33)$$

Using (3.28), we see that (3.33) will hold if

$$\sum_{k=0}^{\infty} \alpha(k+1) \sum_{\ell=0}^k \ell^N < \infty.$$

Note that (3.33) certainly holds if the decay rate $\alpha(k)$ in assumption (C4) is exponential, i.e. if $\alpha(k) = \varkappa^k$ for some $\varkappa \in (0, 1)$. In fact, if we have exponential decay in (C4), then the map $s \mapsto \theta_\omega(s)$ is infinitely differentiable at $s = 0$, and thus, each of the moments of the RV Z exists and is finite.

The next proposition follows from remark 3.16 and elementary calculations.

Proposition 3.17. *If the map $s \mapsto \theta_\omega(s)$ is twice differentiable (with respect to s) at $s = 0$, then, if Z is the compound Poisson RV corresponding to the CF φ coming from theorem 3.14, we have*

$$\mathbb{E}(Z) = \int_{\Omega} t_{\omega} \, d\mathbf{m}(\omega) \quad \text{and} \quad \text{Var}(Z) = \int_{\Omega} t_{\omega} (1 + 2\Sigma_{\omega}) \, d\mathbf{m}(\omega).$$

Proof. Note that if $s \mapsto \theta_\omega(s)$ is twice differentiable, then so is the map $s \mapsto \varphi(s)$. Thus we must have that $D_s \varphi(0) = i\mathbb{E}(Z)$ and $D_s^2 \varphi(0) = i^2 \mathbb{E}(Z^2) = -\mathbb{E}(Z^2)$. Let $u(s) = (e^{is} - 1) \int_{\Omega} t_{\omega} \theta_{\omega}(s) \, d\mathbf{m}(\omega)$, then $\varphi(s) = e^{u(s)}$ by (3.32). Then we have $D_s \varphi(s) = D_s u(s) \cdot \varphi(s)$ and $D_s^2 \varphi(s) = ((D_s u(s))^2 + D_s^2 u(s)) \cdot \varphi(s)$. Calculating $D_s u(s)$ gives

$$D_s u(s) = i e^{is} \int_{\Omega} t_{\omega} \theta_{\omega}(s) \, d\mathbf{m}(\omega) + (e^{is} - 1) \int_{\Omega} t_{\omega} D_s \theta_{\omega}(s) \, d\mathbf{m}(\omega),$$

where (by (3.29))

$$D_s \theta_{\omega}(s) = i e^{is} \sum_{k=0}^{\infty} \sum_{j=0}^k e^{i\ell s} \beta_{\omega,0}^{(k)}(\ell) + (e^{is} - 1) \sum_{k=0}^{\infty} \sum_{\ell=0}^k i \ell e^{i\ell s} \beta_{\omega,0}^{(k)}(\ell).$$

In view of Σ_{ω} defined in (3.27), we note that

$$D_s \theta_{\omega}(0) = i \sum_{k=0}^{\infty} \sum_{j=0}^k \beta_{\omega,0}^{(k)}(\ell) = i \Sigma_{\omega}.$$

Since $\varphi(0) = 1$, $\theta_{\omega}(0) = 1$, and $|D_s \theta_{\omega}(0)| < \infty$ (by (3.28)) we have that

$$D_s \varphi(0) = D_s u(0) = i \int_{\Omega} t_{\omega} \, d\mathbf{m}(\omega),$$

and so $\mathbb{E}(Z) = \int_{\Omega} t_{\omega} \, d\mathbf{m}(\omega)$.

Now calculating $D_s^2 u(s)$ we have

$$D_s^2 u(s) = -e^{is} \int_{\Omega} t_{\omega} \theta_{\omega}(s) \, d\mathbf{m}(\omega) + 2ie^{is} \int_{\Omega} t_{\omega} D_s \theta_{\omega}(s) \, d\mathbf{m}(\omega) + (e^{is} - 1) \int_{\Omega} t_{\omega} D_s^2 \theta_{\omega}(s) \, d\mathbf{m}(\omega).$$

Noting that $|D_s^2 \theta_{\omega}(0)|$ is finite and $\theta_{\omega}(0) = 1$, plugging in $s = 0$ gives that

$$\begin{aligned} D_s^2 \varphi(0) &= (D_s u(0))^2 + D_s^2 u(0) \\ &= i^2 \left(\int_{\Omega} t_{\omega} \, d\mathbf{m}(\omega) \right)^2 + i^2 \int_{\Omega} t_{\omega} \, d\mathbf{m}(\omega) + 2i \int_{\Omega} t_{\omega} D_s \theta_{\omega}(0) \, d\mathbf{m}(\omega) \\ &= - \left(\int_{\Omega} t_{\omega} \, d\mathbf{m}(\omega) \right)^2 - \int_{\Omega} t_{\omega} \, d\mathbf{m}(\omega) - 2 \int_{\Omega} t_{\omega} \Sigma_{\omega} \, d\mathbf{m}(\omega). \end{aligned}$$

Since $D_s^2 \varphi(0) = -\mathbb{E}(Z^2)$, we have

$$\mathbb{E}(Z^2) = \left(\int_{\Omega} t_{\omega} \, d\mathbf{m}(\omega) \right)^2 + \int_{\Omega} t_{\omega} \, d\mathbf{m}(\omega) + 2 \int_{\Omega} t_{\omega} \Sigma_{\omega} \, d\mathbf{m}(\omega). \quad (3.34)$$

Calculating the variance $\text{Var}(Z) = \mathbb{E}(Z^2) - (\mathbb{E}(Z))^2$ finishes the proof. $\square$

Since Z a compound Poisson random variable there exists N , a Poisson RZ with parameter ϑ , and iid RVs X_j such that

$$Z := \sum_{j=1}^N X_j.$$

Furthermore, the CF $\varphi(s) = \phi_Z(s)$ of the RV Z is related to the CF $\phi_{X_i}(s)$ of the RV X_i by the following

$$\phi_Z(s) = e^{\vartheta(\phi_{X_1}(s) - 1)}. \quad (3.35)$$

Arguing as in section 2, for each $\ell \geq 1$, we set $S_{\ell} = \sum_{i=1}^{\ell} X_i$ and S_0 the RV equal to 0. Since Z is compound Poisson we can write

$$\mathbb{P}(Z = k) = \sum_{\ell=0}^{\infty} \mathbb{P}(N = \ell) \mathbb{P}(S_{\ell} = k), \quad (3.36)$$

and so, using Wald's equation and proposition 3.17, we have that $\mathbb{E}(Z) = \vartheta \mathbb{E}(X_1) = \int_{\Omega} t_{\omega} \, d\mathbf{m}(\omega)$. Therefore, we must have that

$$\vartheta = \frac{\int_{\Omega} t_{\omega} \, d\mathbf{m}(\omega)}{\mathbb{E}(X_1)}. \quad (3.37)$$

Now we note that it follows from the definition of the RV Z that $\mathbb{P}(Z = 0)$ is given by Gumbel's law (see [5, section 2.4]), and thus

$$\mathbb{P}(Z = 0) = e^{-\int_{\Omega} t_{\omega} \theta_{\omega,0} \, d\mathbf{m}(\omega)}, \quad (3.38)$$

where $\theta_{\omega,0}$ is the usual extremal index defined by (3.30). Setting $k = 0$ in (3.36), the only term which survives is $\ell = 0$ (since $\mathbb{P}(S_0 = 0) = 1$ as $S_0 \equiv 0$), which gives

$$\mathbb{P}(Z = 0) = \mathbb{P}(N = 0) = e^{-\vartheta} = e^{-\frac{\int_{\Omega} t_{\omega} \, d\mathbf{m}(\omega)}{\mathbb{E}(X_1)}}, \quad (3.39)$$

and therefore using (3.37)–(3.39) to solve for $\mathbb{E}(X_1)$ we get

$$\mathbb{E}(X_1) = \frac{\int_{\Omega} t_{\omega} \, \mathbf{d}m(\omega)}{\int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathbf{d}m(\omega)}. \quad (3.40)$$

Note that this relationship generalises the relationship found in section 2 and in [34]. It follows from (3.37) and (3.40) that

$$\vartheta = \int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathbf{d}m(\omega). \quad (3.41)$$

To find the variance of the RV X_i we differentiate the CF $\phi_Z(s)$ and $\phi_{X_1}(s)$ as in proposition 3.17. In particular, we have

$$D_s \phi_Z(s) = \vartheta D_s \phi_{X_1}(s) \cdot \phi_Z(s)$$

and

$$D_s^2 \phi_Z(s) = \left(\vartheta D_s^2 \phi_{X_1}(s) + (\vartheta D_s \phi_{X_1}(s))^2 \right) \cdot \phi_Z(s).$$

Plugging in $s=0$ together with (3.37), the fact that $D_s \phi_{X_1}(0) = i\mathbb{E}(X_1)$, and (3.40) gives that

$$\begin{aligned} -\mathbb{E}(Z^2) &= D_s^2 \phi_Z(0) = D_s^2 \phi_{X_1}(0) \int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathbf{d}m(\omega) + \left(i \int_{\Omega} t_{\omega} \, \mathbf{d}m(\omega) \right)^2 \\ &= D_s^2 \phi_{X_1}(0) \int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathbf{d}m(\omega) - \left(\int_{\Omega} t_{\omega} \, \mathbf{d}m(\omega) \right)^2. \end{aligned}$$

So using (3.34), the formula for $\mathbb{E}(Z^2)$ from the proof of proposition 3.17, we can solve for $\mathbb{E}(X_1^2) = -D_s^2 \phi_{X_1}(0)$ to get

$$\mathbb{E}(X_1^2) = \frac{\int_{\Omega} t_{\omega} (1 + 2\Sigma_{\omega}) \, \mathbf{d}m(\omega)}{\int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathbf{d}m(\omega)},$$

and so we can calculate the variance of X_1 to be

$$\text{Var}(X_1) = \frac{\int_{\Omega} t_{\omega} (1 + 2\Sigma_{\omega}) \, \mathbf{d}m(\omega)}{\int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathbf{d}m(\omega)} - \left(\frac{\int_{\Omega} t_{\omega} \, \mathbf{d}m(\omega)}{\int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathbf{d}m(\omega)} \right)^2.$$

Using (3.32), (3.35), (3.41), and the fact that $\varphi(s) = \phi_Z(s)$, we can solve for the CF $\phi_{X_1}(s)$ of X_1 to get

$$\phi_{X_1}(s) = (e^{is} - 1) \frac{\int_{\Omega} t_{\omega} \theta_{\omega}(s) \, \mathbf{d}m(\omega)}{\int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathbf{d}m(\omega)} + 1.$$

Thus, we have just proved the following proposition.

Proposition 3.18. *The CF of the compound Poisson RV Z can be written as*

$$\varphi(s) = \phi_Z(s) = e^{\vartheta(\phi_{X_1}(s)-1)},$$

where $Z = \sum_{k=1}^N X_k$, N is a Poisson RV with parameter $\vartheta = \int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathrm{d}m(\omega)$, and $X_1, X_2, \dots$ are RV whose CF is given by

$$\phi_{X_1}(s) = (e^{is} - 1) \frac{\int_{\Omega} t_{\omega} \theta_{\omega}(s) \, \mathrm{d}m(\omega)}{\int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathrm{d}m(\omega)} + 1.$$

Moreover, if the map $s \mapsto \theta_{\omega}(s)$ is twice differentiable (with respect to s) at $s=0$, then, the expectation and variance of the RV X_1 is given by

$$\begin{aligned} \mathbb{E}(X_1) &= \frac{\int_{\Omega} t_{\omega} \, \mathrm{d}m(\omega)}{\int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathrm{d}m(\omega)} \quad \text{and} \\ \text{Var}(X_1) &= \frac{\int_{\Omega} t_{\omega} (1 + 2\Sigma_{\omega}) \, \mathrm{d}m(\omega)}{\int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathrm{d}m(\omega)} - \left(\frac{\int_{\Omega} t_{\omega} \, \mathrm{d}m(\omega)}{\int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathrm{d}m(\omega)} \right)^2. \end{aligned}$$

The distribution of X_1 can be obtained by applying the Lévy inversion formula

$$\mathbb{P}(X_1 = k) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T e^{-isk} \left((e^{is} - 1) \frac{\int_{\Omega} t_{\omega} \theta_{\omega}(s) \, \mathrm{d}m(\omega)}{\int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathrm{d}m(\omega)} + 1 \right) \, \mathrm{d}s. \quad (3.42)$$

Following the procedure outlined in section 2 to solve for $\mathbb{P}(X_1)$ by differentiating the PGF $G_Z(s)$ of Z , and noting that $\vartheta = \int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathrm{d}m(\omega)$, we have

$$\mathbb{P}(Z = K) = \frac{e^{-\int_{\Omega} t_{\omega} \theta_{\omega,0} \, \mathrm{d}m(\omega)}}{K!} \left(\sum_{k=0}^K \frac{1}{k!} D_s^K (\tilde{g}_{X_1}^k)(0) \right). \quad (3.43)$$

Thus one could obtain the distribution for X_1 (as an alternative to using (3.42)) by using (3.43) to solve for $\mathbb{P}(X_1 = k) = D_s^k(\tilde{g}_{X_1})(0)/k!$ recursively in terms of $\mathbb{P}(Z = k)$.

4. Checking our general assumptions for a large class of examples

In this section we present an explicit class of random piecewise-monotonic interval maps for which our general assumptions (C1)–(C7) hold. We follow the approach of section 2.5 of [5] adapted to the setting of our current perturbations.

We now suppose that the spaces $\mathcal{J}_{\omega} = [0, 1]$ for each $\omega \in \Omega$ and the maps $T_{\omega} : [0, 1] \rightarrow [0, 1]$ are surjective, finitely-branched, piecewise monotone, nonsingular (with respect to Lebesgue measure Leb), and that there exists $C \geq 1$ such that

$$\text{ess sup}_{\omega} |T'_{\omega}| \leq C \quad \text{and} \quad \text{ess sup}_{\omega} d(T_{\omega}) \leq C, \quad (\text{F1})$$

where $d(T_{\omega}) := \sup_{y \in [0,1]} \#T_{\omega}^{-1}(y)$. We let $\mathcal{Z}_{\omega,0}$ denote the (finite) monotonicity partition of T_{ω} and for each $k \geq 2$ we let $\mathcal{Z}_{\omega,0}^{(k)}$ denote the partition of monotonicity of T_{ω}^k .

(MC) The map $\sigma : \Omega \rightarrow \Omega$ is a homeomorphism, the skew-product map $T : \Omega \times [0, 1] \rightarrow \Omega \times [0, 1]$ is measurable, and $\omega \mapsto T_{\omega}$ has countable range.

Remark 4.1. Under assumption (MC), the family of transfer operator cocycles $\{\mathcal{L}_{\omega,n,s}\}$ satisfies the conditions of theorem 17 [30] (m -continuity and σ a homeomorphism). Note that condition (MC) implies that T satisfies (M1) and the cocycle generated by $\mathcal{L}_0$ satisfies condition (M2).

Recall that the *variation* of $f: [0, 1] \rightarrow \mathbb{R}_+$ on $Z \subset [0, 1]$ be

$$\text{var}_Z(f) = \sup_{x_0 < \dots < x_k, x_j \in Z} \sum_{j=0}^{k-1} |f(x_{j+1}) - f(x_j)|,$$

and $\text{var}(f) := \text{var}_{[0,1]}(f)$. We let $\text{BV} = \text{BV}([0, 1])$ denote the set of complex-valued functions on $[0, 1]$ that have bounded variation. We define the Banach space $\text{BV}_1 \subseteq L^\infty(\text{Leb})$ to be the set of (equivalence classes of) functions of bounded variation on $[0, 1]$, with norm given by

$$\|f\|_{\text{BV}_1} := \inf_{\tilde{f} = f \text{ Leba.e.}} \text{var}(\tilde{f}) + \text{Leb}(|f|).$$

We denote the supremum norm on $L^\infty(\text{Leb})$ by $\|\cdot\|_{\infty,1}$. We set $J_\omega := |T'_\omega|$ and define the random Perron–Frobenius operator, acting on functions in BV

$$P_\omega(f)(x) := \sum_{y \in T_\omega^{-1}(x)} \frac{f(y)}{J_\omega(y)}.$$

The operator P satisfies the well-known property that

$$\int_{[0,1]} P_\omega(f) \, d\text{Leb} = \int_{[0,1]} f \, d\text{Leb} \quad (4.1)$$

for m -a.e. $\omega \in \Omega$ and all $f \in \text{BV}$. Recall from section 3 that $g_0 = \{g_{\omega,0}\}_{\omega \in \Omega}$ and that

$$\mathcal{L}_{\omega,0}(f)(x) := \sum_{y \in T_\omega^{-1}(x)} g_{\omega,0}(y) f(y), \quad f \in \text{BV}.$$

We assume that the weight function $g_{\omega,0}$ lies in BV for each $\omega \in \Omega$ and satisfies

$$\text{ess sup}_\omega \|g_{\omega,0}\|_{\infty,1} < \infty, \quad (\text{F2})$$

and

$$\text{ess inf}_\omega g_{\omega,0} > 0. \quad (\text{F3})$$

Note that (F1) and (F2) together imply

$$\text{ess sup}_\omega \|\mathcal{L}_{\omega,0} \mathbf{1}\|_{\infty,1} \leq \text{ess sup}_\omega d(T_\omega) \|g_{\omega,0}\|_{\infty,1} < \infty \quad (4.2)$$

and

$$\text{ess sup}_\omega \|g_{\omega,0} J_\omega\|_{\infty,1} < \infty. \quad (4.3)$$

We also assume a uniform covering condition¹³:

(F4) For every subinterval $J \subset [0, 1]$ there is a $k = k(J)$ such that for a.e. ω one has $T_\omega^k(J) = [0, 1]$.

¹³ We could replace the covering condition with the assumption of a strongly contracting potential. See [4] for details.

Concerning the holes system we assume that $H_{\omega,n} \subseteq [0, 1]$ are chosen so that assumption (A) holds. We also assume for each $\omega \in \Omega$ and each $n \geq 1$ that $H_{\omega,n}$ is composed of a finite union of intervals such that

(F5) There is a uniform-in- n and uniform-in- ω upper bound $\mathfrak{h} \geq 1$ on the number of connected components of $H_{\omega,n}$.

We now assume that

$$\lim_{n \rightarrow \infty} \operatorname{ess\,sup}_{\omega} \operatorname{Leb}(H_{\omega,n}) = 0. \quad (\text{F6})$$

Because we are considering small targets shrinking to zero measure, it is natural to assume that

$$T_{\omega}(H_{\omega,n}^c) = [0, 1] \quad (\text{F7})$$

for m -a.e. $\omega \in \Omega$ and all $n \geq 1$ sufficiently large. Further we suppose that there exists $N' \geq 1$ such that ¹⁴

$$(9 + 12\mathfrak{h}N') \cdot \operatorname{ess\,sup}_{\omega} \|g_{\omega,0}^{(N')}\|_{\infty,1} < \operatorname{ess\,inf}_{\omega} \inf \mathcal{L}_{\omega,0}^{N'} \mathbb{1}. \quad (\text{F8})$$

Remark 4.2. Note that if $\operatorname{ess\,sup}_{\omega} \|g_{\omega,0}\|_{\infty,1} < \operatorname{ess\,inf}_{\omega} \inf \mathcal{L}_{\omega,0} \mathbb{1}$, then the gap between $\operatorname{ess\,sup}_{\omega} \|g_{\omega,0}\|_{\infty,1}$ and $\operatorname{ess\,inf}_{\omega} \inf \mathcal{L}_{\omega,0} \mathbb{1}$ grows exponentially, and thus such an N' exists.

For each $k \in \mathbb{N}$ and $\omega \in \Omega$ we let $\mathcal{A}_{\omega,0}^{(k)}$ be the collection of all finite partitions of $[0, 1]$ such that

$$\operatorname{var}_{A_i} \left(g_{\omega,0}^{(k)} \right) \leq 2 \|g_{\omega,0}^{(k)}\|_{\infty,1} \quad (4.4)$$

for each $\mathcal{A} = \{A_i\} \in \mathcal{A}_{\omega,0}^{(k)}$. Given $\mathcal{A} \in \mathcal{A}_{\omega,0}^{(k)}$, let $\mathcal{Z}_{\omega,*}^{(k)}$ be the coarsest partition amongst all those finer than $\mathcal{A}$ and $\mathcal{Z}_{\omega,0}^{(k)}$.

Remark 4.3. Note that if $\operatorname{var}_Z(g_{\omega,0}^{(k)}) \leq 2 \|g_{\omega,0}^{(k)}\|_{\infty,1}$ for each $Z \in \mathcal{Z}_{\omega,0}^{(k)}$ then we can take the partition $\mathcal{A} = \mathcal{Z}_{\omega,0}^{(k)}$. Furthermore, the 2 above can be replaced by some $\hat{\alpha} \geq 0$ (depending on $g_{\omega,0}$) following the techniques of [3, 5].

We assume the following covering condition

(F9) There exists $k_o(N') \in \mathbb{N}$ such that for m -a.e. $\omega \in \Omega$, and all $Z \in \mathcal{Z}_{\omega,*}^{(N')}$ we have $T_{\omega}^{k_o(N')}(Z) = [0, 1]$, where N' is the number coming from (F8).

Remark 4.4. Note that the uniform covering time assumption (F9) clearly holds if (F4) holds and if there are only finitely many maps T_{ω} . Remark 2.C.2 of [5] presents an alternative assumption to (F9) which could be adapted to the setting of this paper.

¹⁴ Note that the coefficients appearing in (F8) are not optimal. See [3, 5] for how this assumption may be improved.

Since $T_\omega^j|_Z$ is one-to-one for each $Z \in \mathcal{Z}_{\omega,*}^{(k)}$ and each $1 \leq j \leq k$, the number of connected components of $Z \cap T_\omega^{-j}(H_{\sigma^j \omega})$ is bounded above by $\mathfrak{h}$ by assumption (F5). Since the function $\exp(\mathrm{i}s1_{H_{\sigma^j \omega,n}} \circ T_\omega^j)$ has at most $2\mathfrak{h}$ jump discontinuities on each interval $Z \in \mathcal{Z}_{\omega,*}^{(k)}$ for each $1 \leq j \leq k$, we have that $e^{\mathrm{i}sS_{\omega,n,k}}$ has at most $2\mathfrak{h}k$ many jump discontinuities. Furthermore, $e^{\mathrm{i}sS_{\omega,n,k}}$ is constant between consecutive discontinuities. Using this together with the fact that the distance in $\mathbb{C}$ between points on the unit circle is bounded above by 2, we have that

$$\mathrm{var}_Z(e^{\mathrm{i}sS_{\omega,n,k}}) \leq 4\mathfrak{h}k$$

for each $Z \in \mathcal{Z}_{\omega,*}^{(k)}$. Recalling that $g_{\omega,n,s}^{(k)} := \exp(S_k(\varphi_{\omega,0}) + \mathrm{i}sS_{\omega,n,k}) = g_{\omega,0} \exp(\mathrm{i}sS_{\omega,n,k})$, where $g_{\omega,0} = \exp(\varphi_{\omega,0})$ is as in section 3, we see that (4.4) implies that

$$\begin{aligned} \mathrm{var}_Z(g_{\omega,n,s}^{(k)}) &\leq \|g_{\omega,0}^{(k)}\|_{\infty,1} \mathrm{var}_Z(e^{\mathrm{i}sS_{\omega,n,k}}) + \|e^{\mathrm{i}sS_{\omega,n,k}}\|_{\infty} \mathrm{var}_Z(g_{\omega,0}^{(k)}) \\ &\leq 4\mathfrak{h}k \|g_{\omega,0}^{(k)}\|_{\infty,1} + 2 \|g_{\omega,0}^{(k)}\|_{\infty,1} \leq \|g_{\omega,0}^{(k)}\|_{\infty,1} (2 + 4\mathfrak{h}k) \end{aligned} \quad (4.5)$$

for each $Z \in \mathcal{Z}_{\omega,*}^{(k)}$.

The following lemma of [5] extends several results in [23] from the specific weight $g_{\omega,0} = 1/|T'_\omega|$ to general weights satisfying the conditions just outlined.

Lemma 4.5 (Lemma 2.5.9 of [5]). *Assume that a family of random piecewise-monotonic interval maps $\{T_\omega\}$ satisfies (F1), (F2), (F3), and (F4), as well as (F8) and (MC) for the unperturbed operator cocycle. Then (CCM) and (C1) hold, and the parts of (C2)–(C5) and (C7) corresponding to the unperturbed operator cocycle hold as well. Further, $\nu_{\omega,0}$ is fully supported and condition (C4) holds with $C_f = K$, for some $K < \infty$, and $\alpha(N) = \gamma^N$ for some $\gamma < 1$.*

From lemma 4.5 we have that $\lambda_{\omega,0} := \nu_{\sigma\omega,0}(\mathcal{L}_{\omega,0}\mathbb{1})$ and thus for each $k \geq 1$ we have

$$\mathrm{ess\,inf}_\omega \lambda_{\omega,0}^k \geq \mathrm{ess\,inf}_\omega \inf \mathcal{L}_{\omega,0}^k \mathbb{1} \geq \mathrm{ess\,inf}_\omega g_{\omega,0}^{(k)} > 0. \quad (4.6)$$

Furthermore, since $\nu_{\omega,0}$ is fully supported on $[0, 1]$ and non-atomic for a.e. ω , we have that (F6) implies that for m -a.e. $\omega \in \Omega \lim_{n \rightarrow \infty} \nu_{\omega,0}(H_{\omega,n}) = 0$, and since $\mathrm{ess\,sup}_\omega \lambda_{\omega,0}$ is bounded (by (4.2)) together with remark 3.4, we have that (C6) holds.

We now use hyperbolicity of the unperturbed transfer operator cocycle to guarantee that we have hyperbolic cocycles for large n , which will yield (C2)–(C6) for large n .

The main result of this section is the following.

Lemma 4.6. *Assume that a family of random piecewise-monotonic interval maps $\{T_\omega\}$ satisfies (F1)–(F9) together with (MC). Then (CCM), (A), and (B) hold as well as (C1). Furthermore, for all $n \geq 1$ sufficiently large, conditions (C2)–(C7) hold. In particular $(\Omega, m, \sigma, [0, 1], T, \mathrm{BV}_1, \mathcal{L}_0, \nu_0, \phi_0, H_n)$ is a random perturbed system.*

Proof. The proof of this lemma follows similarly to the proof 2.5.10 in [5]. The main argument is to apply theorem A of [21]. Denoting $\hat{\mathcal{L}}_{\omega,n,s} := \lambda_{\omega,0}^{-1} \mathcal{L}_{\omega,n,s}$ for each $\omega, n \geq 1$, and $s \in \mathbb{R} \setminus \{0\}$ we can write the hypotheses of theorem A [21], in our notation, as:

- (1) $\hat{\mathcal{L}}_{\omega,0}$ is a hyperbolic transfer operator cocycle on BV_1 with norm $\|\cdot\|_{\mathrm{BV}_1}$ and a one-dimensional leading Oseledets space (see definition 3.1 [21]), and slow and fast growth rates $0 < \gamma < \Gamma$, respectively. We will construct γ and Γ shortly.

(2) The family of cocycles $\{\hat{\mathcal{L}}_{\omega,n,s}\}_{n \geq n_0}$ satisfy a uniform Lasota–Yorke inequality

$$\|\hat{\mathcal{L}}_{\omega,n,s}^k f\|_{BV_1} \leq A\alpha^k \|f\|_{BV_1} + B^k \|f\|_1$$

for a.e. ω and $n \geq n_0$, where $\alpha \leq \gamma < \Gamma \leq B$.

(3) $\lim_{n \rightarrow \infty} \text{ess sup}_{\omega} \|\hat{\mathcal{L}}_{\omega,0} - \hat{\mathcal{L}}_{\omega,n,s}\| = 0$, where $\|\cdot\|$ is the $BV - L^1$ (Leb) triple norm.

The proofs of items (1) and (3) are similar to the proofs given in the proof of lemma 2.5.10 in [5]. To prove item (2), we require the following Lasota–Yorke inequality inspired by lemma 1.5.1 of [5].

Lemma 4.7. *For any $f \in BV$ we have*

$$\text{var}(\mathcal{L}_{\omega,n,s}^k(f)) \leq (9 + 12\mathfrak{h}k) \|g_{\omega,0}^{(k)}\|_{\infty,1} \text{var}(f) + \frac{(8 + 12\mathfrak{h}k) \|g_{\omega,0}^{(k)}\|_{\infty,1}}{\min_{Z \in \mathcal{Z}_{\omega,*}^{(k)}} \nu_{\omega,0}(Z)} \nu_{\omega,0}(|f|).$$

Proof. By considering intervals Z in $\mathcal{Z}_{\omega,*}^{(k)}$, we are able to write

$$\mathcal{L}_{\omega,n,s}^k f = \sum_{Z \in \mathcal{Z}_{\omega,*}^{(k)}} \left(\mathbb{1}_Z f g_{\omega,n,s}^{(k)} \right) \circ T_{\omega,Z}^{-k} \quad (4.7)$$

where

$$T_{\omega,Z}^{-k} : T_{\omega}^k(I_{\omega,n,s}) \rightarrow Z$$

is the inverse branch that takes $T_{\omega}^k(x)$ to x for each $x \in Z$. Now, since

$$\mathbb{1}_Z \circ T_{\omega,Z}^{-k} = \mathbb{1}_{T_{\omega}^k(Z)},$$

we can rewrite (4.7) as

$$\mathcal{L}_{\omega,n,s}^k f = \sum_{Z \in \mathcal{Z}_{\omega,*}^{(k)}} \mathbb{1}_{T_{\omega}^k(Z)} \left(\left(f g_{\omega,n,s}^{(k)} \right) \circ T_{\omega,Z}^{-k} \right). \quad (4.8)$$

So,

$$\text{var}(\mathcal{L}_{\omega,n,s}^k f) \leq \sum_{Z \in \mathcal{Z}_{\omega,*}^{(k)}} \text{var} \left(\mathbb{1}_{T_{\omega}^k(Z)} \left(\left(f g_{\omega,n,s}^{(k)} \right) \circ T_{\omega,Z}^{-k} \right) \right). \quad (4.9)$$

Now for each $Z \in \mathcal{Z}_{\omega,*}^{(k)}$, using (4.5), we have

$$\begin{aligned} \text{var} \left(\mathbb{1}_{T_{\omega}^k(Z)} \left(\left(f g_{\omega,n,s}^{(k)} \right) \circ T_{\omega,Z}^{-k} \right) \right) &\leq \text{var}_Z \left(f g_{\omega,n,s}^{(k)} \right) + 2 \sup_Z \left| f g_{\omega,n,s}^{(k)} \right| \\ &\leq 3 \text{var}_Z \left(f g_{\omega,n,s}^{(k)} \right) + 2 \inf_Z \left| f g_{\omega,n,s}^{(k)} \right| \\ &\leq 3 \|g_{\omega,n,s}^{(k)}\|_{\infty,1} \text{var}_Z(f) + 3 \sup_Z |f| \text{var}_Z \left(g_{\omega,n,s}^{(k)} \right) + 2 \|g_{\omega,n,s}^{(k)}\|_{\infty,1} \inf_Z |f| \\ &\leq 3 \|g_{\omega,0}^{(k)}\|_{\infty,1} \text{var}_Z(f) + (6 + 12\mathfrak{h}k) \|g_{\omega,0}^{(k)}\|_{\infty,1} \sup_Z |f| + 2 \|g_{\omega,0}^{(k)}\|_{\infty,1} \inf_Z |f| \end{aligned}$$

$$\begin{aligned}
&\leq (9 + 12\mathfrak{h}k) \|g_{\omega,0}^{(k)}\|_{\infty,1} \text{var}_Z(f) + (8 + 12\mathfrak{h}k) \|g_{\omega,0}^{(k)}\|_{\infty,1} \inf_Z |f| \\
&\leq (9 + 12\mathfrak{h}k) \|g_{\omega,0}^{(k)}\|_{\infty,1} \text{var}_Z(f) + (8 + 12\mathfrak{h}k) \|g_{\omega,0}^{(k)}\|_{\infty,1} \frac{\nu_{\omega,0}(|f|Z)}{\nu_{\omega,0}(Z)}. \quad (4.10)
\end{aligned}$$

Using (4.10), we may further estimate (4.9) as

$$\begin{aligned}
\text{var}(\mathcal{L}_{\omega,n,s}^k f) &\leq \sum_{Z \in \mathcal{Z}_{\omega,*}^{(k)}} \left((9 + 12\mathfrak{h}k) \|g_{\omega,0}^{(k)}\|_{\infty,1} \text{var}_Z(f) + (8 + 12\mathfrak{h}k) \|g_{\omega,0}^{(k)}\|_{\infty,1} \frac{\nu_{\omega,0}(|f|Z)}{\nu_{\omega,0}(Z)} \right) \\
&\leq (9 + 12\mathfrak{h}k) \|g_{\omega,0}^{(k)}\|_{\infty,1} \text{var}(f) + \frac{(8 + 12\mathfrak{h}k) \|g_{\omega,0}^{(k)}\|_{\infty,1}}{\min_{Z \in \mathcal{Z}_{\omega,*}^{(k)}} \nu_{\omega,0}(Z)} \nu_{\omega,0}(|f|), \quad (4.11)
\end{aligned}$$

and thus the proof of lemma 4.7 is complete. $\square$

To complete the proof of item (2) of lemma 4.6 we begin by setting

$$\alpha^{N'} := \frac{(9 + 12\mathfrak{h}N') \text{ess sup}_{\omega} \|g_{\omega,0}^{(N')}\|_{\infty,1}}{\text{ess inf}_{\omega} \inf \mathcal{L}_{\omega,0}^{N'} \mathbb{1}} < 1, \quad (4.12)$$

which is possible by (F8). For item (2) using the the Lasota–Yorke inequality proven in lemma 4.7, together with (4.12) and dividing through by $\lambda_{\omega,0}^{N'}$ gives

$$\begin{aligned}
\text{var}(\hat{\mathcal{L}}_{\omega,n,s}^{N'} f) &\leq \frac{(9 + 12\mathfrak{h}N') \|g_{\omega,0}^{(N')}\|_{\infty,1}}{\lambda_{\omega,0}^{N'}} \text{var}(f) + \frac{(8 + 12\mathfrak{h}N') \|g_{\omega,0}^{(N')}\|_{\infty,1}}{\lambda_{\omega,0}^{N'} \min_{Z \in \mathcal{Z}_{\omega,*}^{(N')}(\mathcal{A})} \nu_{\omega,0}(Z)} \nu_{\omega,0}(|f|) \\
&\leq \alpha^{N'} \text{var}(f) + \frac{\alpha^{N'}}{\min_{Z \in \mathcal{Z}_{\omega,*}^{(N')}(\mathcal{A})} \nu_{\omega,0}(Z)} \nu_{\omega,0}(|f|). \quad (4.13)
\end{aligned}$$

We note that (F3), (F9), (4.6), and the equivariance of the backward adjoint cocycle imply that for $Z \in \mathcal{Z}_{\omega,*}^{(N')}$ we have

$$\nu_{\omega,0}(Z) = \nu_{\sigma^{k_0(N')} \omega,0} \left(\left(\lambda_{\omega,0}^{k_0(N')} \right)^{-1} \mathcal{L}_{\omega,0}^{k_0(N')} \mathbb{1}_Z \right) \geq \frac{\inf g_{\omega,0}^{k_0(N')}}{\lambda_{\omega,0}^{k_0(N')}} > 0. \quad (4.14)$$

As stated above, the remainder of the proof of lemma 4.6 follows similarly to the proof of lemma 2.5.10 in [5]. $\square$

5. Examples

In this section we present several examples of random dynamics resulting in varying types of compound Poisson distributions. We provide examples of systems in the class described in section 4.

Example 5.1. Compound Poisson that is not Pólya–Aeppli from random maps and random targets: Let $\{\Omega_j\}$ be a partition of Ω . We consider the following family of maps $\{T_{\omega}\}$:

$$T_{\omega}(x) = \begin{cases} L_{\omega}(x), & 0 \leq x \leq (1 - 1/\gamma_{\omega})/2; \\ \gamma_{\omega}x - (\gamma_{\omega} - 1)/2, & (1 - 1/\gamma_{\omega})/2 \leq x \leq (1 + 1/\gamma_{\omega})/2; \\ R_{\omega}(x), & (1 + 1/\gamma_{\omega})/2 \leq x \leq 1, \end{cases} \quad (5.1)$$

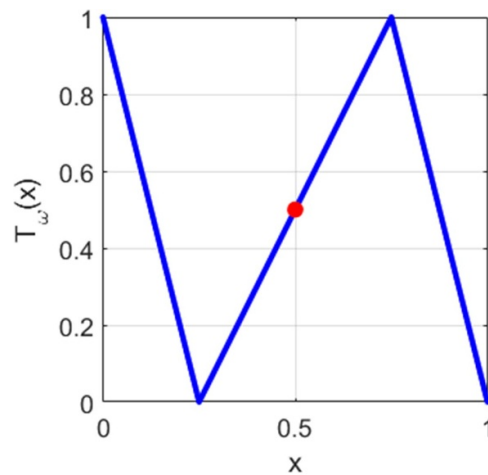


Figure 1. Graph of a map T_ω , with $\gamma_\omega = 2$.

where $L|_{\Omega_j}$ and $R|_{\Omega_j}$ are maps with finitely many full linear branches, $1 < \gamma \leq \gamma_\omega \leq \Gamma < \infty$ and $\gamma|_{\Omega_j}$ is constant for each $j \geq 1$. We take $g_{\omega,0} = 1/|DT_\omega|$ and since these maps have full linear branches, Lebesgue measure is preserved by all maps T_ω ; i.e. $\mu_{\omega,0} = \text{Leb}$ for a.e. ω . The central branch has slope γ_ω and passes through the fixed point $x_0 = 1/2$, which lies at the centre of the central branch; see figure 1. A specific random driving could be $\sigma : S^1 \rightarrow S^1$ given by $\sigma(\omega) = \omega + \varrho$ for some $\varrho \notin \mathbb{Q}$ and $\gamma_\omega = \gamma^0 + \gamma^1 \cdot \omega$ for $1 < \gamma^0 < \infty$ and $0 < \gamma^1 < \infty$, but only the ergodicity of σ will be important for us. We consider the Banach spaces $\mathcal{B}_\omega$ to be the space BV of complex-valued functions of bounded variation.

For each $\omega \in \Omega$ and each $n \in \mathbb{N}$ consider a sequence of shrinking balls $H_{\omega,n}$ containing the fixed point x_0 such that $x_0 \in H_{\omega,n+1} \subseteq H_{\omega,n}$ for each $n \in \mathbb{N}$. Then assumption (B) is clearly satisfied. Furthermore the targets $H_{\omega,n}$ are chosen so that the assumption (S) is satisfied. Such targets $H_{\omega,n}$ are constructed in section 2.7 of [5] as neighbourhoods of extremal points of observation functions $h : M \rightarrow \mathbb{R}$.

To verify our general assumptions (M1), (M2), (CCM), (A), (C1)–(C7) it suffices to check the assumptions (F1)–(F9)¹⁵. This is done in a similar manner to the way in which the assumptions (E1)–(E9) of [5, section 2.5] are checked in [5, example 2.7.1] for transfer operators acting on the space of real-valued BV functions with different perturbations. Therefore we have only to verify assumptions (C8) and (S).

Assumption (S) states that $\mu_{\omega,0}(H_{\omega,n}) \leq \frac{|t|_\infty + W}{n}$. In this example, all fibre measures $\mu_{\omega,0}$ are Lebesgue and so the targets $H_{\omega,n}$ are neighbourhoods of x_0 of diameter no greater than $(|t|_\infty + W)/n$. Because the maps T_ω are locally expanding about x_0 , for fixed $k > 0$ one can find a large enough n so that it is impossible to leave a small neighbourhood of x_0 and return after k iterations. Therefore we can always find n sufficiently large to guarantee that $\beta_{\omega,n}^{(k)}(\ell) = 0$ for all $0 \leq \ell < k$. If this were not the case, there must be a positive $\mu_{\sigma^{-(k+1)}\omega,0}$ -measure set of points that (i) lie in $H_{\sigma^{-(k+1)}\omega,n}$, (ii) land inside $\ell < k$ of the targets $H_{\sigma^{-k}\omega,n}, \dots, H_{\sigma^{-1}\omega,n}$ over the course of the next k iterations, and then (iii) land in $H_{\omega,n}$ on the $k+1$ st iteration, which we have argued is impossible. On the other hand, the equality $\ell = k$ corresponds to the situation

¹⁵ We note that since the targets consist of a single interval, to to check (F8) with $N' = 1$, it suffices to ensure that $\text{ess inf}_\omega \gamma_\omega > 21$ and that the slopes of the branches of the maps L_ω and R_ω are at least as large as γ_ω .

where points remain in the sequence of targets for *all* k iterations, which will occur for a positive measure set of points. This immediately implies that $\beta_{\omega,n}^{(k)}(k) \neq 0$, and thus

$$\hat{q}_{\omega,n}^{(k)}(s) = (1 - e^{is}) e^{iks} \beta_{\omega,n}^{(k)}(k) = (1 - e^{is}) e^{iks} \frac{\mu_{\sigma^{-(k+1)}\omega,0} \left(\bigcap_{j=0}^{k+1} T_{\sigma^{-(k+1)}\omega}^{-j} (H_{\sigma^{-(k+1)+j}\omega,n}) \right)}{\mu_{\omega,0}(H_{\omega,n})}.$$

Because the scaling t_ω varies along the orbit $\sigma^{-(k+1)}\omega, \dots, \omega$, to simplify the above expression we assume that $\text{ess sup}_\omega t_{\sigma\omega}/t_\omega < \text{ess inf}_\omega |T'_\omega(x_0)|$. This mild assumption on the variation of the scaling t removes the ‘Case 2’ considerations in example 2.7.1 [5], and taking the limit as $n \rightarrow \infty$ gives

$$\hat{q}_{\omega,0}^{(k)}(s) = (1 - e^{is}) \frac{e^{iks}}{|DT_{\sigma^{-(k+1)}\omega}^{k+1}(x_0)|} = (1 - e^{is}) e^{iks} \prod_{j=0}^k \alpha_{\sigma^{-(k+1)+j}\omega},$$

where we have set $\alpha_\omega = \gamma_\omega^{-1} = |DT_\omega(x_0)|^{-1}$, and thus we have

$$\theta_\omega(s) = 1 - (1 - e^{is}) \sum_{k=0}^{\infty} \frac{e^{iks}}{|DT_{\sigma^{-(k+1)}\omega}^{k+1}(x_0)|}.$$

Applying theorem 3.14 gives that

$$\begin{aligned} \varphi(s) &= \exp \left(- (1 - e^{is}) \int_{\Omega} t_\omega \left(1 - (1 - e^{is}) \sum_{k=0}^{\infty} \frac{e^{iks}}{|DT_{\sigma^{-(k+1)}\omega}^{k+1}(x_0)|} \right) dm(\omega) \right) \\ &= \exp \left(- (1 - e^{is}) \int_{\Omega} t_\omega \left(1 - (1 - e^{is}) \sum_{k=0}^{\infty} e^{iks} \prod_{j=0}^k \alpha_{\sigma^{-(k+1)+j}\omega} \right) dm(\omega) \right). \end{aligned}$$

Note that since $1 < \gamma \leq \gamma_\omega \leq \Gamma$, and so $1 > \alpha_\omega \geq \Gamma^{-1} > 0$, the series $\sum_{k=0}^{\infty} e^{iks} \prod_{j=0}^k \alpha_{\sigma^{-(k+1)+j}\omega}$ converges absolutely to say $\Sigma_\omega(s)$ (which depends on ω). Since $\Sigma_\omega(s)$ is not necessarily a geometric series, then $\varphi(s)$ is the characteristic function of a compound Poisson distribution which is not the Pólya–Aeppli distribution. It is unclear which specific compound Poisson distribution is represented by $\varphi(s)$.

Note that the tail of the series $\Sigma_\omega(s)$ is approximately given by $e^{-is} \sum_{k=N_\omega}^{\infty} (e^{is} \int_{\Omega} \log \alpha_\omega dm)^{k+1}$ where N_ω is the (ω -dependent) time that it takes for the Birkhoff Ergodic Theorem to apply.

Example 5.2. Pólya–Aeppli from non-random slope and random targets: Taking the family of maps $\{T_\omega\}$ from the previous example, we now suppose that the reciprocal of the slope of the central branch is constant; i.e. $1/DT_\omega(x_0) = \alpha_\omega \equiv \alpha > 1$ is constant. Then we have

$$\begin{aligned} \varphi(s) &= \exp \left(- (1 - e^{is}) \int_{\Omega} t_\omega \left(1 - (1 - e^{is}) \sum_{k=0}^{\infty} e^{iks} \alpha^{k+1} \right) dm(\omega) \right) \\ &= \exp \left(- (1 - \alpha) \left(\int_{\Omega} t_\omega dm(\omega) \right) \left(\frac{1 - e^{is}}{1 - e^{is}\alpha} \right) \right). \end{aligned}$$

Then $\varphi(s)$ is the characteristic function of a Pólya–Aeppli (geometric Poisson) distributed random variable Z with parameters given by $\rho = \alpha \in (0, 1)$ and $\vartheta = (1 - \alpha) \int_{\Omega} t_\omega dm$.

Example 5.3. Pólya–Aeppli from I.I.D. Maps and targets: Now suppose that we are in the setting of the previous example where now we have two maps T_1 and T_2 chosen iid with probabilities p_1 and p_2 respectively. Again we let x_0 denote the common fixed point of the maps T_1 and T_2 and let α_i denote the reciprocal of the slope of the map T_i at x_0 , i.e. $\alpha_i = 1/DT_i(x_0)$, for $i = 1, 2$. Further suppose there are two values of t , namely t_1 and t_2 , which are chosen iid with probabilities ϱ_1 and ϱ_2 respectively¹⁶. Such a model can be implemented by $\sigma : \{0, 1\}^{\mathbb{Z}} \times \{0, 1\}^{\mathbb{Z}}$, where $\sigma\omega := \sigma((\omega^{(T)}, \omega^{(t)})) := (\sigma^{(T)}\omega^{(T)}, \sigma^{(t)}\omega^{(t)})$, with $\sigma^{(T)}$ the left shift acting on the sequence $\omega^{(T)}$ controlling the choice of map and $\sigma^{(t)}$ the left shift acting on the sequence $\omega^{(t)}$ controlling the choice of scaling constant t . We set $m = m^{(T)} \times m^{(t)}$ where $m^{(T)}$ is the Bernoulli measure generated by (p_1, p_2) and $m^{(t)}$ is the Bernoulli measure generated by (ϱ_1, ϱ_2) . Thus, using the fact that the α are chosen iid, we have that

$$\int_{\Omega} \prod_{j=0}^k \alpha_{\sigma^{-(k+1)+j}\omega} dm(\omega) = (p_1\alpha_1 + p_2\alpha_2)^{k+1} =: \bar{\alpha}^{k+1} \quad (5.2)$$

Note that $\bar{\alpha} \in (0, 1)$. Now since t_{ω} is independent of the α terms in the product appearing in (5.2) (since the product does not contain a α_{ω}), we can use (5.2) to write that

$$\begin{aligned} \int_{\Omega} t_{\omega} \left(1 - (1 - e^{is}) \sum_{k=0}^{\infty} e^{iks} \prod_{j=0}^k \alpha_{\sigma^{-(k+1)+j}\omega} \right) dm(\omega) \\ = \bar{t} \left(1 - (1 - e^{is}) \sum_{k=0}^{\infty} e^{iks} \int_{\Omega} \prod_{j=0}^k \alpha_{\sigma^{-(k+1)+j}\omega} dm(\omega) \right) \\ = \bar{t} \left(1 - (1 - e^{is}) \sum_{k=0}^{\infty} e^{iks} \bar{\alpha}^{k+1} \right) = \bar{t} \left(\frac{1 - \bar{\alpha}}{1 - e^{is}\bar{\alpha}} \right), \end{aligned}$$

where $\bar{t} = \int_{\Omega} t_{\omega} dm = t_1\varrho_1 + t_2\varrho_2$. Inserting this into the formula for $\varphi(s)$ gives

$$\varphi(s) = \exp \left(-\bar{t} (1 - e^{is}) \left(\frac{1 - \bar{\alpha}}{1 - e^{is}\bar{\alpha}} \right) \right) = \exp \left(-\bar{t} (1 - \bar{\alpha}) \left(\frac{1 - e^{is}}{1 - e^{is}\bar{\alpha}} \right) \right),$$

which is the characteristic function of a Pólya–Aeppli ($\rho = \bar{\alpha} \in (0, 1)$ and $\vartheta = (1 - \bar{\alpha})\bar{t}$) distributed random variable.

Example 5.4. Standard Poisson from random maps and random targets: Unlike the previous examples, we now provide an example in which the characteristic function produced from theorem 3.14 is for a standard Poisson random variable rather than compound Poisson. We recall the setting of example 4 of [5]. Let $\Omega = \{0, 1, 2, 3\}^{\mathbb{Z}}$, with σ the bilateral shift map, and m an invariant ergodic measure.

¹⁶ The argument here goes through exactly the same if t takes on countably many values t_i with probability ϱ_i . Furthermore, the argument still goes through if the t values are chosen according to some ergodic process rather than being chosen i.i.d. In this case example 5.2 can be seen as a special case of example 5.3.

To each letter $j = 0, \dots, l-1$ we associate a rational number $v_j \in (0, 1) \cap \mathbb{Q}$. We consider targets $H_{\omega, n} = B(v_{\omega_0}, e^{-z_n(\omega)})$, where ω_0 denotes the 0-th coordinate of $\omega \in \Omega$ and the thresholds $z_n(\omega)$ are chosen such that assumption (S) holds. For each $\omega \in \Omega$ we associate a map T_{ω_0} , where $T_0, \dots, T_3$ are maps of the circle which we will take as β -maps of the form $T_i(x) = \beta_i x + r \pmod{1}$, with $\beta_i \in \mathbb{N}$, $\beta_i \geq 3$, and $0 \leq r < 1$ irrational and independent of i . Similar arguments to those given in [5, example 4] show that our assumptions (M1), (M2), (CCM), (A), (C1)–(C7) are satisfied for this system. This is accomplished as in example 5.1, by showing that our assumptions (F1)–(F9) follow from arguments similar to those of [5, example 4] used to verify the assumptions (E1)–(E9). Furthermore, assumption (B) is satisfied by taking the Banach spaces $\mathcal{B}_\omega$ again to be BV.

Following the argument of [5, example 4], it follows that a necessary condition to get $\beta_{\omega, n}^{(k)}(\ell) \neq 0$ for some $k \geq 0$ and $0 \leq \ell \leq k$, and thus $\hat{q}_{\omega, 0}^{(k)}(s) \neq 0$, is that the centre $v_{(\sigma^{-(k+1)}\omega)_0}$ is sent to the centre v_{ω_0} . Let z be one of these rational centres. The iterate $T_\omega^n(z)$ has the form $T_\omega^n(z) = \beta_{\omega_{n-1}} \cdots \beta_{\omega_0} z + k_n r \pmod{1}$, where k_n is an integer. Therefore such an iterate will never be a rational number, which shows that all $\hat{q}_{\omega, 0}^{(k)}(s) = 0$ for each $k \geq 0$, ω , and each $s \in \mathbb{R} \setminus \{0\}$. Therefore we have $\theta_\omega(s) \equiv 1$, and so applying theorem 3.14, we have

$$\varphi(s) = \exp\left(-\left(1 - e^{is}\right) \int_{\Omega} t_\omega dm(\omega)\right),$$

which is the characteristic function of a Poisson random variable with parameter $\vartheta = \int_{\Omega} t_\omega dm(\omega)$.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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